

The Wannier-Functions Software Ecosystem for Materials Simulations

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Over the last two decades, following the early developments on maximally-localized Wannier functions, an ecosystem of electronic-structure simulation techniques and software leveraging the Wannier representation has flourished. This environment includes codes to obtain Wannier functions and interfaces with first-principles simulation software, as well as an increasing number of related post-processing packages. Wannier functions can be obtained for isolated or extended systems (both crystalline and disordered), and can be used to understand chemical bonding, to characterize polarization, magnetization, and topology, or as an optimal basis set, providing very accurate interpolations in reciprocal space or large-scale Hamiltonians in real space. In this review, we summarize the current landscape of techniques, materials properties and simulation codes based on Wannier functions that have been made accessible to the research community, and that are now well integrated into what we term a *Wannier-functions software ecosystem*. First we introduce the theory and practicalities of Wannier functions, starting from their broad domains of applicability to advanced minimization methods using alternative approaches beyond maximal localization. Then we define the concept of a Wannier ecosystem and its interactions and interoperability with many quantum engines and post-processing packages. We focus on some of the key properties that are empowered by such ecosystem—from band interpolations and large-scale simulations to electronic transport, Berryology, topology, electron-phonon couplings, dynamical mean-field theory, embedding, and Koopmans functionals—concluding with the current status of interoperability and automation. The review aims at highlighting basic theory and concepts behind codes, providing relevant pointers to more in-depth references. It also elucidates the relationships and connections between codes and, where relevant, the different motivations and objectives behind their development strategies. Finally, we provide an outlook on future developments, and comment on the goals of biodiversity and sustainability for the whole software ecosystem.

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I. Introduction

Wannier functions (WFs) (Wannier, 1937), and in particular maximally-localized Wannier functions (MLWFs) (Marzari and Vanderbilt, 1997), provide an accurate, compact, and localized representation of the electronic-structure problem, and have become widely used in computational condensed-matter physics and materials science (Marzari *et al.*, 2012).

Thanks to developments in theory, algorithms and implementations over the past few decades, summarized in Sec. II, it has now become possible to apply widely the concept of MLWFs to single-particle theories and in particular to Kohn–Sham (KS) density-functional theory (DFT) simulations, to obtain localized orbitals from Bloch states; the latter can be themselves represented with localized or extended basis sets, such as plane waves. On one hand, these developments have benefited from profound connections between WFs and physical quantities such as electric polarization, orbital magnetization and topological invariants (Soluyanov and Vanderbilt, 2011b; Thonhauser *et al.*, 2005; Vanderbilt, 2018; Vanderbilt and King-Smith, 1993; Xiao *et al.*, 2005). On the other hand, the ability to obtain MLWFs from DFT simulations often allows to calculate physical quantities with high accuracy, but at a fraction of the computational cost, thanks to their role as very accurate interpolators (Lee *et al.*, 2005; Souza *et al.*, 2001; Yates *et al.*, 2007). Finally, although not discussed here, localized representations have long been pioneered by the quantum chemistry community to interpret coordination and bonding (Edmiston and Ruedenberg, 1963), and MLWFs extend to periodic systems the concept of Foster-Boys localized orbitals (Boys, 1966), thanks to algorithmic breakthroughs in calculating the position operator in solids (Blount, 1962; King-Smith and Vanderbilt, 1993; Nenciu, 1991; Resta, 1992; Zak, 1989).

Wannier functions are typically localized or even exponentially localized (Brouder *et al.*, 2007; Panati, 2007; Panati and Pisante, 2013), and due to the nearsightedness of interacting electrons (Des Cloizeaux, 1964a,b; Kohn, 1996), local electronic properties only depend on the nearby environment (Bianco and Resta, 2011, 2013; Marrazzo and Resta, 2016, 2019). As a consequence,

the resulting Hamiltonian matrix expressed in a localized basis set (such as MLWFs (Calzolari *et al.*, 2004; Lee *et al.*, 2005)) becomes sparse, i.e., it displays negligible matrix elements—or hoppings, in the language of a tight-binding (TB) formalism—if the distance between the corresponding localized basis functions exceeds a given threshold. In this sense, MLWFs are optimal choices as they decay exponentially in real space (Panati and Pisante, 2013) and they minimize a localization functional by design (Marzari *et al.*, 2012; Marzari and Vanderbilt, 1997). The resulting MLWFs can be used as a basis set to build, LEGOTM-like, the electronic structure of large-scale nanostructures (Lee *et al.*, 2005) that in turn could be solved with linear-scaling methods (Mauri *et al.*, 1993; Nunes and Vanderbilt, 1994; Ordejón *et al.*, 1993), or as a remarkably accurate interpolators of electronic properties, operators and quantities defined as integrals over the Brillouin zone (BZ) of periodic systems (Souza *et al.*, 2001; Yates *et al.*, 2007). Interpolation on dense grids becomes essential when very fine features need to be resolved, as happens when integrals are restricted to lower-dimensional manifolds (such as the Fermi surface, in the case of transport properties of metals).

Nowadays, MLWFs are routinely used in many research areas of condensed-matter physics and materials science. In Sec. II we summarize the past and current challenges, discussing how we reached the current state. Such a flourishing is not only due to theoretical advances, but also strongly driven by the concerted development of accessible and efficient software. Indeed, thanks to the availability of robust software packages (often open-source, encouraging further contributions), and to the user support provided by developers, researchers can now not only easily compute MLWFs, but also use them as core ingredients for advanced simulations. As more codes appear, they adopt the *de facto* standardization of input and output formats, resulting in a set of interacting and interoperating codes that we will call here the “Wannier-functions software ecosystem.”

This review does not aim to provide an extensive discussion of the theory of MLWFs, for which we refer to Marzari *et al.* (2012), although we do provide a general introduction to the field in Sec. I.A. Instead, the goal is to discuss the nature of the ecosystem and the capabilities of existing codes, focusing in Sec. III on a selection of physical phenomena or quantities that can be efficiently predicted thanks to WFs, and on how WFs are used as an ingredient to extend the accuracy of beyond-DFT simulations. Nevertheless, we will still mention a few notable developments of the past decade, whenever useful to contextualize the theoretical and software developments. Our aim is to provide a reference that can help newcomers and existing practitioners alike navigate the ecosystem: which properties can be computed by which codes, why the use of WFs is beneficial, which quantities are exchanged between codes, and how interoperability

is being addressed.

To discuss the codes, we group them into three major categories: *Wannier engines*, i.e., codes to obtain WFs; *interface codes* between the first-principles engines (e.g., DFT or GW codes) and the Wannier engines; and *Wannier-enabled codes*, which range from relatively simple post-processing tools to more advanced codes that use WFs as one of the ingredients to accelerate accurate simulations. A fourth category of codes that we discuss in Sec. III.J are *automation workflows*. Indeed, until very recently the generation of WFs typically required human intuition by experienced researchers to provide initial trial orbitals. This barrier has been largely removed by recent algorithmic and automation efforts (see Sec. II.D), enabling the use of WFs both by new users and for high-throughput materials discovery and characterization. In the latter case, managing thousands (or more) simulations poses new challenges, which require not only the use of robust workflow engines, but also the implementation of WF-specific workflows to effectively interconnect multiple codes within the ecosystem.

We will finally conclude in Sec. IV with some perspectives on the field related to the sustainability of the whole effort, current challenges that still need to be addressed, and possible future developments.

A. Ab initio electronic structure and Wannier functions

Electronic-structure simulations aim to determine the behavior of electrons in materials and molecules, as governed by the Schrödinger or Dirac equations. Electrons feel Coulomb interactions among themselves and with the nuclei, in addition to couplings with external fields (e.g., electrical, magnetic, or electromagnetic/photons) or perturbations (e.g., strain, phonons). The core electrons of heavy chemical elements can reach relativistic speeds, requiring to correct the Schrödinger equation with terms obtained from an expansion of the Dirac equation in powers of $1/c^2$, where c is the speed of light. The most relevant relativistic correction is spin-orbit coupling, which is responsible for several important phenomena related to magnetism and to geometrical and topological properties of the electronic manifold (see Sec. III.E and Sec. III.F).

An exact solution of the Schrödinger equation (either the bare one, or with relativistic corrections) would give access to essentially all property of materials. This problem was already clear in 1929, when Paul Dirac declared: “The underlying physical laws necessary for the mathematical theory of a large part of physics and the whole of chemistry are thus completely known, and the difficulty is only that the exact application of these laws leads to equations much too complicated to be soluble” (Dirac, 1929).

For more than fifty years, *ab initio* or first-principles methods have been developed to approximately solve the

Schrödinger equation in realistic settings (Giustino, 2014; Martin, 2020), with more accurate strategies and theories being developed. In parallel, the exponential growth of computational power (Moore’s law) has allowed to deploy these new theoretical instruments through numerical simulations, constantly seeking not only to improve accuracy but also targeting more complex and realistic systems. Quickly, numerical solutions became sufficiently accurate to be predictive for a number of properties in relevant systems: the era of first-principles materials modeling had begun (Marzari *et al.*, 2021; Yin and Cohen, 1980).

An iconic example is given by DFT, which made it possible to determine the electronic structure of complex materials with reasonable accuracy at low cost. In DFT, the total energy of the electrons is expressed as a functional of the electronic charge density (Giustino, 2014; Martin, 2020). The theory is supported by two pillars, the Hohenberg–Kohn theorems, which state not only a one-to-one correspondence between the ground-state many-body wavefunction and the ground-state charge density, but also formulate a variational total-energy functional: the solution of the Schrödinger equation can be recast as a minimization problem for the charge density, a remarkably simpler object (a real function of $\mathbf{r}$) than the many-body wavefunction we started from (a complex function of $3N_e$ variables, where N_e is the number of electrons in the system).

In KS DFT, the interacting many-body problem is mapped onto a non-interacting problem sharing the same ground-state charge density but in presence of a suitable local external potential; the latter is in general unknown. This *ansatz* allows to capture with accuracy the kinetic energy contribution through the second derivatives of the KS orbitals, otherwise invisible in practice in their square moduli (the density). The success of DFT has been possible also thanks to the discovery and development of simple functionals that approximate the exact, but unknown, total-energy functional. Hence, the KS-DFT approach to solving the many-body Schrödinger equation translates into solving a set of non-interacting one-particle Schrödinger equations in presence of an external potential that depends, self-consistently, upon the charge density only. Although the outstanding importance of DFT has been recognized by the 1998 Nobel Prize in chemistry awarded to Walter Kohn (for DFT) and John Pople (for computational methods in quantum chemistry), its impact on the physics community has been, if possible, even greater: the top ten most highly cited articles published by the American Physical Society deal with DFT and related applications (Talirz *et al.*, 2021).

The electronic structure of materials and molecular systems is, at the same time, very similar and yet very different. To some extent, extended materials can be seen as very large molecules and one could focus on the local electronic structure in real space, which is periodi-

cally repeated in the case of a crystalline materials (e.g., metals, semiconductors, oxides). This viewpoint is supported by the mathematical structure of the Schrödinger equation and its solutions, as observed by Kohn (Kohn, 1996): the electronic structure is fundamentally a local property, “nearsighted” to what happens at further distance in real space. The effect of chemical bonding and the presence of the lattice can be seen as perturbations to the case of isolated atoms, with their well-defined s , p , d and f orbitals. This perspective is powerful and foundational for linear-scaling methods, which target the simulation of large systems and leverage a description based on localized orbitals. Yet materials are not just very large molecules, and “more is different” (Anderson, 1972). Extended systems are practically infinite and can thus be described using periodic boundary conditions (PBCs). As will be discussed in more formal terms in Sec. II, the electronic structure of a material in PBC is more naturally described in terms of Bloch orbitals, which are not localized in real space. Indeed, a different perspective often drives the discussion of the electronic structure of periodic crystals: the behavior in reciprocal (i.e., Fourier-transformed) space. A textbook example is semiconductor physics, that is in general much better understood (and taught) by studying solutions of the Schrödinger equation in such reciprocal space. This approach, somewhat orthogonal to the large-molecule perspective, is indeed also very powerful both at a conceptual and practical level. As a side note, many electronic-structure codes for materials actually adopt a completely delocalized basis (e.g., plane waves) to describe the periodic part of the Bloch orbitals (Martin, 2004, 2020; Pickett, 1989).

How can we reconcile these two, almost opposite, perspectives? Remarkably, one can use the fundamental “gauge freedom” of quantum mechanics: First, any quantum state is defined modulo a phase factor. Second, if one considers a set of single-particle states separated in energy from other states, then any trace operation on this manifold is invariant with respect to any unitary transformation among the orbitals; we call this a “generalized” gauge freedom. However, and this is the crucial aspect, the localization properties of that set of states strongly depend on their gauge.

WFs provide a rigorous and insightful way to reconcile the real-space and localized perspective with the reciprocal-space (Fourier) and delocalized one. As it will be clear in the next section, MLWFs in particular exploit the generalized gauge freedom to transform delocalized orbitals into localized ones (and vice versa), by constructing the proper unitary matrices.

II. Wannier functions fundamentals

The electronic structure of periodic crystals is most commonly described in terms of Bloch waves

$\psi_{n\mathbf{k}}(\mathbf{r}) = u_{n\mathbf{k}}(\mathbf{r}) e^{i\mathbf{k}\cdot\mathbf{r}}$, where $\mathbf{k}$ is the crystal momentum, and n is the band index. This is the case in textbooks on solid-state theory, but also for the many software packages that solve the KS equations for crystalline solids. A few years after Felix Bloch developed the theory of electron waves in periodic crystals (Bloch, 1929), Gregory Wannier introduced an alternative representation in terms of an orthonormal set of localized functions (Wannier, 1937), the Wannier functions. Given an isolated Bloch band n , the WF $w_{n\mathbf{R}}(\mathbf{r}) = \langle \mathbf{r} | \mathbf{R}n \rangle = w_{n0}(\mathbf{r} - \mathbf{R})$ associated with the unit cell labelled by lattice vector $\mathbf{R}$ is defined as (Wannier, 1937)

$$|\mathbf{R}n\rangle = \frac{V_{\text{cell}}}{(2\pi)^3} \int_{\text{BZ}} d\mathbf{k} e^{-i\mathbf{k}\cdot\mathbf{R}} |\psi_{n\mathbf{k}}\rangle, \quad (1)$$

where V_{cell} is the unit-cell volume.

Since their inception, WFs have been employed as a conceptual tool to tackle problems in solid-state physics (see, e.g., Kivelson (1982)). However, in the first sixty years following Wannier’s paper, there were few actual calculations of WFs for real materials (see, e.g., Callaway and Hughes (1967); Satpathy and Pawłowska (1988); and Sporkmann and Bross (1994)). The main obstacle was the fact that WFs are strongly non-unique, being sensitive to the generalized gauge freedom discussed earlier, e.g., to k -dependent phase changes $|\psi_{n\mathbf{k}}\rangle \rightarrow e^{-i\beta_{n\mathbf{k}}} |\psi_{n\mathbf{k}}\rangle$ with $\beta_{n\mathbf{k}} \in \mathbb{R}$ (gauge transformations) of the Bloch eigenstates. Moreover, the energy bands of real materials typically become degenerate at points, lines, or even entire planes in the BZ. The presence of degeneracies leads to poor localization properties of the WFs obtained from Eq. (1), because no matter how the phase factors $e^{-i\beta_{n\mathbf{k}}}$ are chosen, the Bloch eigenfunctions are non-differentiable functions of $\mathbf{k}$ at the degeneracy points.

A. Maximally-localized Wannier functions

In the following, we discuss the original Wannierization methods introduced in Marzari and Vanderbilt (1997) and Souza *et al.* (2001), while we refer the reader to Sec. II.D for an overview of more recent and advanced minimization methods.

1. Isolated composite groups of bands

Consider a group of J Bloch bands of orthonormal $|\psi_{n\mathbf{k}}\rangle$ Bloch states that may be connected among themselves by degeneracies, but are isolated from all lower or higher bands, e.g., the six valence bands in Fig. 1. Given such a composite group, the most general expression for

the associated WFs is (Marzari and Vanderbilt, 1997)

$$|\mathbf{R}_j\rangle = \frac{V_{\text{cell}}}{(2\pi)^3} \int_{\text{BZ}} d\mathbf{k} e^{-i\mathbf{k}\cdot\mathbf{R}} |\psi_{j\mathbf{k}}^{\text{W}}\rangle, \quad (2a)$$

$$|\psi_{j\mathbf{k}}^{\text{W}}\rangle = \sum_{n=1}^J |\psi_{n\mathbf{k}}\rangle U_{\mathbf{k},nj}, \quad (2b)$$

where the $U_{\mathbf{k}}$ are $J \times J$ unitary matrices that describe the generalized (multiband) gauge freedom within the Bloch manifold at each $\mathbf{k}$. The superscript W denotes a Wannier gauge, as opposed to a Hamiltonian gauge (later denoted by H) where the Hamiltonian matrix is diagonal. Note that at variance with Eq. (1), in Eq. (2) there is not a one-to-one correspondence between the band index n and the intra-cell Wannier index j .

Marzari and Vanderbilt (MV) introduced the concept of MLWFs, in which the $U_{\mathbf{k}}$ matrices are chosen so as to minimize the total quadratic spread of the WFs (Marzari and Vanderbilt, 1997):

$$\Omega = \sum_{j=1}^J \left[\langle \mathbf{0}j | r^2 | \mathbf{0}j \rangle - |\langle \mathbf{0}j | \mathbf{r} | \mathbf{0}j \rangle|^2 \right]. \quad (3)$$

As discussed later (see Sec. II.C.1) the spread (a.k.a. localization) functional Ω and its gradient with respect to an infinitesimal gauge transformation can be expressed in reciprocal space; furthermore, the BZ integration in Eq. (2a) is replaced by a discrete sum $(1/N) \sum_{\mathbf{k}}$ where N is the number of k -points in the finite grid used in the numerical simulations, and the optimal $U_{\mathbf{k}}$ matrices are found by iteratively minimizing the functional Ω (see also Marzari and Vanderbilt (1997) for the mathematical details).

From general Fourier-transform considerations (Duffin, 1953), the good real-space localization properties of the MLWFs on the left-hand side of Eq. (2a) mean that the Bloch-like states $|\psi_{j\mathbf{k}}^{\text{W}}\rangle$ appearing on the right-hand side are smooth functions of $\mathbf{k}$ for the optimal choice of $U_{\mathbf{k}}$ matrices in Eq. (2b) (or for any other choice leading to well-localized WFs).

The details of the MV methodology can be found in Marzari *et al.* (2012) and Marzari and Vanderbilt (1997); in the case of single k -point sampling (large unit cells), it is equivalent to the Foster-Boys scheme used in quantum chemistry to construct localized molecular orbitals (Boys, 1966). It should be noted that other localization criteria can be used for the purpose of obtaining localized orbitals, e.g., the Edmiston–Ruedenberg (Edmiston and Ruedenberg, 1963) and Pipek–Mezey (Pipek and Mezey, 1989) approaches, based on maximizing the Coulomb self-repulsion of the orbitals and the sum of the squares of the Mulliken charges (Mulliken, 2004) associated with the orbitals, respectively. Whilst these are more challenging to adapt to a periodic, multi- k -point formulation, there has been recent work to obtain

WFs for periodic systems using the Pipek–Mezey localization criterion (Clement *et al.*, 2021; Jónsson *et al.*, 2017). Nevertheless, the MV approach of minimizing the quadratic spread is still the most widely used approach for periodic systems.

2. Entangled bands

The MV approach described above provides a means to construct well-localized WFs from isolated groups of bands, such as the valence bands of insulators. However, it is often useful to obtain WFs from non-isolated (or “entangled”) groups of bands. Typical examples include the low-lying conduction bands or the valence plus conduction bands of insulators (see Fig. 1), and the bands crossing the Fermi level in metals.

A possible strategy to deal with such cases is to first identify an appropriate J -dimensional Bloch manifold at each k -point from a larger set of $\mathcal{J}_{\mathbf{k}}$ Bloch eigenstates $|\psi_{m\mathbf{k}}\rangle$, e.g., the ones within some energy window. Formally, this band-disentanglement step can be expressed as

$$|\tilde{\psi}_{n\mathbf{k}}\rangle = \sum_{m=1}^{\mathcal{J}_{\mathbf{k}}} |\psi_{m\mathbf{k}}\rangle \tilde{V}_{\mathbf{k},mn}, \quad (4)$$

where the $\tilde{V}_{\mathbf{k}}$ are $\mathcal{J}_{\mathbf{k}} \times J$ matrices satisfying $\tilde{V}_{\mathbf{k}}^\dagger \tilde{V}_{\mathbf{k}} = \mathbb{1}_{J \times J}$. In 2001, Souza, Marzari and Vanderbilt (Souza *et al.*, 2001) (SMV) introduced a practical scheme to extract an optimally-smooth Bloch-like subspace $\hat{P} = \sum_{n=1}^J |\tilde{u}_{n\mathbf{k}}\rangle \langle \tilde{u}_{n\mathbf{k}}|$ across the BZ, from which a set of MLWFs could then be obtained using the MV prescription. The resulting “disentangled WFs” are given by Eq. (2) with the *ab initio* Bloch eigenstates $|\psi_{n\mathbf{k}}\rangle$ therein replaced by $|\tilde{\psi}_{n\mathbf{k}}\rangle$, that is,

$$|\mathbf{R}_j\rangle = \frac{1}{N} \sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{R}} |\psi_{j\mathbf{k}}^{\text{W}}\rangle, \quad (5a)$$

$$|\psi_{j\mathbf{k}}^{\text{W}}\rangle = \sum_{n=1}^{\mathcal{J}_{\mathbf{k}}} |\psi_{n\mathbf{k}}\rangle V_{\mathbf{k},nj}, \quad (5b)$$

where the $\mathcal{J}_{\mathbf{k}} \times J$ matrices $V_{\mathbf{k}} = \tilde{V}_{\mathbf{k}} U_{\mathbf{k}}$ encode the net result of the disentanglement (subspace-selection) and maximal localization (gauge-selection) steps. As in the case of Eq. (2), the states $|\psi_{j\mathbf{k}}^{\text{W}}\rangle$ in Eq. (5) are smooth functions of $\mathbf{k}$ whenever the associated WFs are well localized.

The disentanglement step can be carried out in such a way that the *ab initio* eigenstates are described exactly within a “frozen” or “inner” energy window that is contained by the “outer” energy window mentioned earlier (Souza *et al.*, 2001). This is useful, for example, when studying transport properties, for which one would like to obtain a faithful description of the states within some

small energy range around the Fermi level. Note that because of these energy windows, the needed input from the first-principles calculation includes the energy eigenvalues $\varepsilon_{n\mathbf{k}}$ in addition to the overlap matrices Eq. (8).

Over the years, many alternative approaches and algorithms have been developed—from partially-occupied Wannier functions (Fontana *et al.*, 2021; Thygesen *et al.*, 2005) to quasiatomic orbitals (Qian *et al.*, 2008), to the selected columns of the density matrix (SCDM) (Damle *et al.*, 2015, 2017), and to projectability disentanglement and manifold remixing (Qiao *et al.*, 2023a,b); these and others will be discussed in Sec. II.D.

B. Major applications of Wannier functions

a. Interpolation The efficient interpolation in reciprocal space of k -dependent quantities is arguably the most common application of WFs, enabling the calculation of simple (e.g., the band structure) or complex (e.g., electron-phonon coupling) electronic structure properties. A large part of this review is devoted to the fundamentals of WF interpolation (Sec. III.C) and their applications, including ballistic transport (Sec. III.D), Berry-phase related properties (Sec. III.E) and electron-phonon interactions (Sec. III.G). As discussed in more detail in Sec. III.C, the reason for such widespread set of applications (not all of them covered in this review) is that WFs can be applied to any generic operator that is local in reciprocal space. Equally important is that WFs allow to reproduce the correct band connectivity: in particular, avoided crossings are not mistaken for actual crossings. This distinguishes Wannier interpolation from other methods based on direct Fourier interpolation of the energy eigenvalues. In other words, WFs allow to exploit the fundamental locality (“nearsightedness” according to Kohn (Des Cloizeaux, 1964a,b; Kohn, 1996)) of the electronic structure and the related exponential localization of WFs to construct a potentially exact representation of an operator in real space, such that any interpolation back to reciprocal space is exact as well. The procedure is also systematic as WFs are guaranteed to exist and the convergence is exponential with the linear sampling density (Brouder *et al.*, 2007; Panati, 2007; Panati and Pisante, 2013); prefactors and coefficients might depend on electronic structure properties such as the band gap, and on the specific operator under consideration.

b. Geometry and Topology WFs have several profound connections with quantum-geometrical and topological aspects of the electronic structure (Vanderbilt, 2018); some of them are discussed in Sec. III.E and III.F. In the following, we refer to topological properties as a subset of geometrical properties that are quantized and

hence represented by integer topological invariants robust to certain classes of perturbations. A prime example of geometrical—and in some circumstances also topological—quantity is the electric polarization of periodic solids, which can be calculated in reciprocal space as a Berry phase (Vanderbilt, 2018) (see Sec. III.E). Electric polarization can also be equivalently computed by summing over WFs centers in real space (Marzari *et al.*, 2012; Marzari and Vanderbilt, 1997; Vanderbilt, 2018), which provides a more intuitive formulation of the modern theory of polarization (King-Smith and Vanderbilt, 1993; Resta, 1992, 1994; Vanderbilt, 2018) and restores some justification to the classical Clausius–Mossotti (Clausius, 1879; Mossotti, 1850) viewpoint. While electronic-structure geometry in reciprocal space speaks the language of differential geometry (e.g., curvatures, parallel transport, smoothness of manifolds), WFs allow to express the same quantities in terms of matrix elements of the Hamiltonian and position operator $\hat{\mathbf{r}}$. The reciprocal-space smoothness, which is measured by the quantum geometric tensor (Provost and Vallee, 1980), can be equivalently analyzed in real space by measuring the degree of WFs localization.

To some extent, the connection between WFs and topological invariants is even stronger, where the former provide not only powerful approaches to calculate invariants for real materials (see Sec. III.F for a discussion) but also fundamental understanding of topological phases. In fact, topological insulators are essentially systems that cannot be connected adiabatically to atomic insulators, hence it is impossible to truly represent their ground state with WFs (Thonhauser and Vanderbilt, 2006; Vanderbilt, 2018). **In reciprocal space, non-trivial topological invariants translate into unavoidable obstructions to choosing a smooth gauge over the BZ** (Bernevig and Hughes, 2013; Vanderbilt, 2018). The fundamental connection between non-trivial topology of electronic bands and the corresponding absence of a Wannier representation has been recently generalized and made systematic in the context of elementary band representation (Michel and Zak, 1999; Zak, 1982), leading to the so-called topological quantum chemistry (Bradlyn *et al.*, 2017; Cano *et al.*, 2018; Vergniory *et al.*, 2017) (see also related efforts on symmetry-based indicators by Khalaf *et al.* (2018); Kruthoff *et al.* (2017); Po *et al.* (2017); and Song *et al.* (2018)), allowing to screen materials databases and identify non-trivial materials of various classes (Tang *et al.*, 2019; Vergniory *et al.*, 2019; Wieder *et al.*, 2022; Zhang *et al.*, 2019).

c. Advanced electronic-structure methods DFT simulations of periodic solids can be conveniently (but definitely not necessarily) performed by adopting a plane-wave basis set, in conjunction with smooth pseudopotentials that reproduce the interaction between valence electrons and

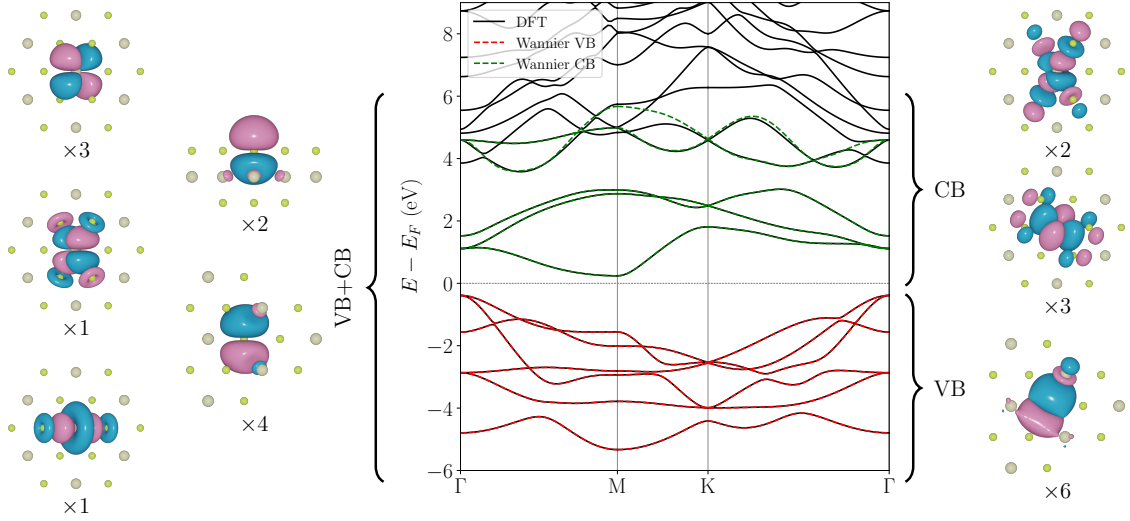


FIG. 1 MLWFs and band structure of the 2D material HfSe₂. In the center, the comparison between the DFT band structure (black lines) and the Wannier-interpolated band structure from valence MLWFs only (VB, red dashed lines) and from low-lying conduction MLWFs only (CB, green dashed lines) are shown. Note that the Wannier-interpolated bands from both valence + conduction MLWFs together (VB+CB) are not shown since they are visually indistinguishable from the combination of the VB and CB MLWFs. On the left, we show the real-space shapes of (in total 11) valence + conduction MLWFs, obtained starting from Hf *d* and Se *p* initial guess orbitals, followed by disentanglement (see Sec. II.A.2) from high-energy conduction states: specifically, three of them resemble $d_{xy,xz,yz}$ orbitals, one resembles a d_{z^2} orbital, one resembles a $d_{x^2-y^2}$ orbital, and the remaining six resemble *p* orbitals. Some small hybridization with orbitals from nearby atoms is visible. On the right, we show the real-space shapes of (in total 6) valence MLWFs (lower panel) and (in total 5) conduction MLWFs (upper panel). The valence MLWFs span an isolated group of bands (see Sec. II.A.1) and are composed by six hybridized bonding orbitals, where the Hf *d* and Se *p* orbitals overlap constructively. The conduction MLWFs are instead five hybridized anti-bonding orbitals, where Hf *d* and Se *p* orbitals overlap destructively, forming nodal planes close to bond centers. (The notation $\times n$ below each shape denotes the multiplicity of the corresponding MLWF, i.e., *n* MLWFs having similar shapes but different spatial orientation.)

nuclei plus core electrons (Martin, 2020). The resulting KS eigenstates are also not particularly localized functions, and DFT is invariant under unitary rotations of the occupied electronic states. However several electronic-structure methods, aiming at improving or complementing the capabilities of DFT, fundamentally require to be formulated in terms of localized orbitals (see Sec. III.H). In addition, several of these beyond-DFT methods are not deployed directly on the crystal structure, but operate more as corrections to starting DFT calculations. Also, beyond-DFT methods can be computational rather intensive, and it is common practice to apply them only on a subset of bands extracted from the entire manifold. In this context, WFs provide a robust way to bridge DFT with advanced electronic-structure methods by allowing to systematically construct orthogonal localized states that represent the manifold of interest: WFs are first constructed on the KS DFT solution and then fed into beyond-DFT methods; a technical overview of how this is carried out in practice is the subject of Sec. III.H.

C. Wannier functions for the practitioner

1. The spread functional in reciprocal space

The Blount identities (Blount, 1962) provide the matrix elements of the position operator between WFs, and, remarkably, prefigure the link between macroscopic properties and integrals (Berry phases) of Berry connections (King-Smith and Vanderbilt, 1993):

$$\langle \mathbf{R}i | \mathbf{r} | \mathbf{0}j \rangle = i \frac{V_{cell}}{(2\pi)^3} \int d\mathbf{k} e^{i\mathbf{k} \cdot \mathbf{R}} \langle u_{i\mathbf{k}} | \nabla_{\mathbf{k}} | u_{j\mathbf{k}} \rangle \quad (6)$$

and

$$\langle \mathbf{R}i | r^2 | \mathbf{0}j \rangle = -\frac{V_{cell}}{(2\pi)^3} \int d\mathbf{k} e^{i\mathbf{k} \cdot \mathbf{R}} \langle u_{i\mathbf{k}} | \nabla_{\mathbf{k}}^2 | u_{j\mathbf{k}} \rangle. \quad (7)$$

It is through these identities that one can recast the spread functional Ω using reciprocal-space expressions, where the gradients and higher derivatives are obtained from finite differences. The building blocks for these finite-difference expressions are the overlap matrices

$$M_{ij}^{(\mathbf{k}, \mathbf{b})} = \langle u_{i\mathbf{k}} | u_{j\mathbf{k}+\mathbf{b}} \rangle \quad (8)$$

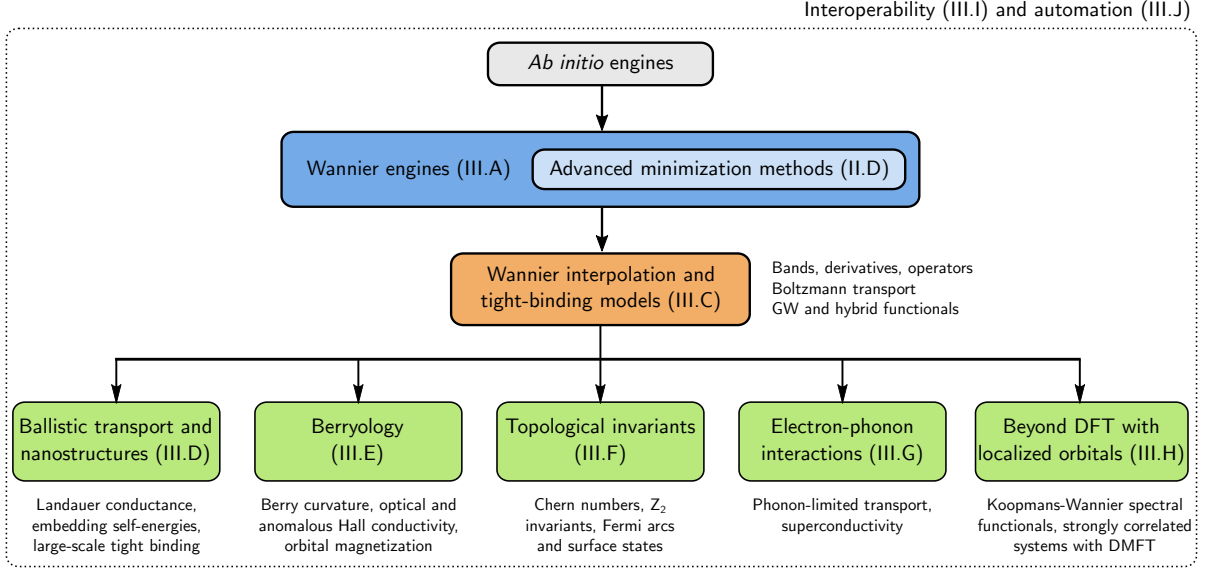


FIG. 2 Overall schematics of how different codes in the ecosystem interact. This figure serves also as a “table of contents” to this review, where sections describing the relevant blocks are indicated in the figure. At the top, the *ab initio* engines (grey block) generate the data that is transferred to a Wannier engine (blue block), that might implement also advanced methods discussed in Sec. II.D. Once WFs are obtained, they are typically used to generate a TB model and perform interpolation of the Hamiltonian and other operators (orange block). Such interpolated quantities are used for a number of different applications; the ones that are discussed in more detail in this review are indicated in the green blocks. As schematically indicated by the outer dotted rectangle, all these codes forming the Wannier ecosystem may be automated with workflow tools, which also coordinate data transfer between them.

between cell-periodic Bloch eigenstates $|u_{n\mathbf{k}}\rangle$ at neighboring points on a regular grid in the BZ (the $\mathbf{b}$ vectors connect one $\mathbf{k}$ -point to its neighbors on a regular discrete grid); in the limit of very dense meshes the $\mathbf{b}$ vectors tend to zero and the gradient in $\mathbf{k}$ is recovered. This finite-difference construction remains valid even in the case of Γ -sampling, where the neighboring $\mathbf{k}$ -points are given by the primitive reciprocal lattice vectors $\mathbf{G}$, with the Bloch orbitals differing just by phase factors $\exp(i\mathbf{G} \cdot \mathbf{r})$. Now, we note that the gradient in $\mathbf{k}$ of a function $f(\mathbf{k})$ can be written as

$$\nabla f(\mathbf{k}) = \sum_{\mathbf{b}} w_b \mathbf{b} [f(\mathbf{k} + \mathbf{b}) - f(\mathbf{k})] + \mathcal{O}(b^2) \quad (9)$$

using stars (“shells”) of neighboring $\mathbf{k}$ -points, where each shell has a weight w_b (see Appendix B in Marzari and Vanderbilt, 1997 and Mostofi *et al.*, 2008 for a detailed description); this is easily proven taking a linear function $f(\mathbf{k}) = f_0 + \mathbf{g} \cdot \mathbf{k}$, where the exact result $\nabla_{\mathbf{k}} f(\mathbf{k}) = \mathbf{g}$ is recovered. In the simple case of cubic Bravais lattices, the first shell of reciprocal-space nearest neighbors (6, 8 or 12 for Bravais lattices in direct space that are simple cubic, FCC, or BCC) is sufficient; the general case where several shells need to be chosen automatically is detailed in Mostofi *et al.* (2008). While the procedure is automated, for unusual cases (e.g., very elongated cells) it might be convenient to find manually the most symmetric choice of shells (Posternak *et al.*, 2002). With these

definitions, the diagonal matrix elements of the position operator can be evaluated by finite differences as

$$\langle 0i|\mathbf{r}|0i\rangle = -\frac{1}{N} \sum_{\mathbf{k}, \mathbf{b}} w_b \mathbf{b} \operatorname{Im} \ln M_{ii}^{(\mathbf{k}, \mathbf{b})}, \quad (10)$$

where N is the number of $\mathbf{k}$ -points of the reciprocal-space grid. More complex expressions for the second derivatives and for the entire spread functional can then be obtained; among many alternatives (all correct to leading order in b), the choices made in Marzari and Vanderbilt (1997) transform correctly under a transformation that shifts $|\mathbf{R}i\rangle$ by a lattice vector. We also note that when using such finite-difference formalism, the spread functional converges slowly (polynomially) with reciprocal-space sampling, and hence some care must be paid in comparing its actual values in case of calculations performed with different discrete samplings.

Importantly, the finite-differences construction is particularly convenient for constructing MLWFs in a code-agnostic form, as the only input needed from the original first-principles calculation is encoded in the overlap matrices $M_{ij}^{(\mathbf{k}, \mathbf{b})}$. Thus, once the $M_{ij}^{(\mathbf{k}, \mathbf{b})}$ have been calculated, no further interaction is necessary with the electronic-structure code that calculated the ground state wavefunctions, making the entire Wannierization procedure a code-independent post-processing step (see, e.g., Ferretti *et al.*, 2007 for the extension to ultrasoft pseudopotentials and the projector-augmented wave method,

and Freimuth *et al.*, 2008; Kune *et al.*, 2010; and Posternak *et al.*, 2002 for the full-potential linearized augmented planewave method). As regards the disentanglement procedure, note that because of energy windows the needed input from the first-principles calculation includes the energy eigenvalues $\varepsilon_{n\mathbf{k}}$ in addition to the overlap matrices Eq. (8).

2. Accuracy and convergence

While the main focus of this review is on the powerful applications of WFs, their successful use relies on the Wannierization process being done correctly: in the following, we briefly comment on fundamental tests and established procedures to assess and improve the quality of WFs.

Two major convergence parameters control the quality (and the cost) of the Wannierization procedure: the spread minimization and the k -point grid used to obtain the initial Hamiltonian eigenstates (e.g., the cell-periodic part of Bloch states, if working with PBCs).

The spread minimization is generally performed with a steepest-descent algorithm until results do not change within a certain tolerance. While the iterative algorithm is in general robust, the minimization can become trapped in local minima. The strategy to avoid that is to select a very good starting point: if the initial spread is sufficiently close to the absolute minimum, it is more likely to reach it by following the local gradient. Hence, particular care needs to be paid to select good projection functions to obtain the initial unitary matrix of Eq. (5b), to be then iteratively optimized. In absence of chemical intuition, a common strategy is to calculate a projected density of states on the pseudoatomic orbitals and identify the orbital character in the energy region of interest: the atomic orbitals which project more on the relevant bands can be used as initial projection. Note that MLWFs are often not atom-centered, and atomic orbitals are not always good starting projections, as it the case of the valence bands of monolayer MoS₂ (Gibertini *et al.*, 2014). In Sections II.D and III.J we cover advanced methods to automate the selection of the starting point for the minimization procedure.

The spread functional measures the degree of localization in real space and, to some extent, the efficiency of the interpolation: more localized WFs decay faster in real space, hence they require a smaller Born-von Kármán (BvK) supercell to include all non-vanishing matrix elements of the Hamiltonian and the other operators, which in turn allows adopting coarser k -point grids in the starting electronic-structure simulation that is performed in reciprocal space. Indeed, the accuracy of band interpolation can be considered a proxy for the quality of the underlying WFs not only as regards k -point convergence: especially in the case of entangled bands (see Sec. II.A.2),

poor interpolation might signal problems in the disentanglement procedure. In addition, it is worth to emphasize that in general the Wannierization procedure is not forced to preserve symmetries (unless dedicated methods designed to do so are employed, see Sec. II.D). Hence, the spurious splitting of symmetry-protected degeneracies in the interpolated band structure might signal convergence problems related to the minimization, to the k -point convergence or to the choice of projection functions. This holds true not only for crystalline symmetries, but also for time-reversal symmetry, which is particularly relevant in non-collinear simulations of non-magnetic materials in presence of spin-orbit coupling (e.g., topological insulators).

Among other indicators of the quality of WFs we mention the ratio between their imaginary (Im) and real (Re) part: for isolated bands (and not considering spin-orbit coupling), MLWFs at the global minimum should be real functions (Marzari and Vanderbilt, 1997). Note that the calculation of the Im/Re part, and anything related to WFs themselves and their visualization, requires to have access to the full Bloch orbitals, not just the overlap matrices. It is also important to emphasize that different quantities derived from WFs—such as the WF spread and centers, as well as unitary matrices $\tilde{V}_{\mathbf{k}}$, $U_{\mathbf{k}}$ —converge in general with different speeds, also depending on the specific formulation adopted (Stengel and Spaldin, 2006).

Finally, we remark that many of the complications related to producing WFs for periodic solids are related to construction of a smooth gauge across the BZ. Hence, supercell Wannierizations with Γ -only sampling are typically more straightforward and less prone to be trapped in local minima. The challenge there is more on algorithmic efficiency due the large size of the systems involved; a number of Γ -only dedicated methods have been developed (Gygi *et al.*, 2003; Silvestrelli, 1999).

D. Advanced minimization methods, and beyond maximally-localized Wannier functions

As discussed in section II, there is in principle large freedom in choosing the recipe to obtain well-localized WFs. Not only one can replace the MV spread functional with other cost functions, but also different minimization procedures and their starting points can be chosen, hence affecting the resulting WFs and their localization properties. Over the years, a number of methods have been developed to address all these different aspects of the Wannierization procedure. It is worth emphasizing that for many of these methods, the initial guess already provides well-localized WFs, so that an iterative minimization can in principle be avoided. The unitary matrices $U_{\mathbf{k}}$ of these “projection-only” WFs are set directly by the initial projection functions. While this choice cannot guarantee optimal localization properties, it has the ad-

vantage of enforcing some degree of symmetry induced by the choice of atomic orbitals used as projection functions. However, in all these cases it is possible—and in some cases even recommended—to minimize the MV spread or some other functional as a final step.

The prime decision deals with the functional that is to be minimized in order to determine the unitary matrices of Eq. (5). The most popular choice is the MV MLWF procedure (Marzari and Vanderbilt, 1997) for composite bands (see Sec. II.A.1) and the SMV disentanglement scheme for entangled bands (Souza *et al.*, 2001) (see Sec. II.A.2). The minimization of the spread functional leads to very well localized WFs, hence reducing the size of the BvK supercell needed to represent operators (e.g., the Hamiltonian) in a WF basis. While MLWFs and disentanglement represent the most convenient choice in the vast majority of applications, substantial work has been done to augment the MLWF scheme or develop alternatives which satisfy the needs of specific applications.

The MLWF procedure leads to optimal real-space localization, but is not guaranteed to yield orbitals that preserve desirable crystal symmetries. This is only partially relieved using symmetric initial projections, as typically obtained by a proper selection of atomic-like orbitals (we further discuss initial projections later on in this section). Symmetry-preserving WFs are appealing in providing the correct orbital or site symmetries for many-body approaches like dynamical mean-field theory (DMFT). Hence, it is not surprising that several non-MLWF procedures directly or indirectly include crystal symmetries in the functional to minimize. In the symmetry-adapted Wannier function (SAWF) method (Sakuma, 2013), symmetric WFs are obtained through additional constraints on the unitary matrices $U_{\mathbf{k}}$ which are based on symmetry operations of the site-symmetry group. The SAWF method is fully compatible with the maximal-localization procedure and the SMV disentanglement (and has also been recently extended to the case where a frozen window is used (Koretsune, 2023)), although the additional constraints imply a possibly larger total spread, even if some individual WFs can actually be more localized than in the MLWF procedure. Currently, the implementation of the SAWF method in the code **Wannier90** (more about software in Sec. III.A) is interfaced with the **Quantum ESPRESSO** distribution and, if the site-symmetry group is equivalent to the full space group of the crystal, then symmetry operations can be calculated directly by the interface code. Otherwise, if the site-symmetry group contains fewer operations than the full space group, then symmetry operations can be provided by an external file. We note that it is also necessary to specify the site location and orbital symmetry of the SAWFs through the projection functions.

While the SAWF method provides a rigorous way to include symmetries in the maximal-localization procedure, it requires quite some prior knowledge of the electronic

structure of the material under study. An alternative and simpler approach is to construct selectively-localized Wannier functions (SLWFs), where the MLWF procedure is applied only to a subset of the entire WFs considered (Wang *et al.*, 2014). In addition, some WF centers can be constrained (SLWF+C) to specific positions by adding a quadratic penalty term to the spread functional. While the SLWF+C approach does not enforce symmetries, it has been observed that the resulting WFs typically exhibit the site symmetries corresponding to the constrained centers (Wang *et al.*, 2014). The SLWF+C can be used in the case of entangled bands, where the SMV disentanglement step is performed as usual, while the selective localization and constrained centers are applied only to the final Wannierization step. A more in-depth review of the SAWF and SLWF+C methods, including their implementation and usage in **Wannier90**, can be found in Pizzi *et al.* (2020).

The localization and possibly the symmetry can crucially depend on the number of WFs considered in a given energy range. The so-called partly occupied Wannier functions (POWFs) (Thygesen *et al.*, 2005) formalize this observation by including the relevant unoccupied states which lead to the minimal spread functional, essentially implementing a bonding-antibonding closing procedure. POWFs can have a high degree of symmetry, while the bonding-antibonding criterion has been shown to correspond to the condition of maximal average localization (Thygesen *et al.*, 2005). Notably, in the POWF scheme the total spread functional $\Omega = \Omega_I + \tilde{\Omega}$ is minimized at once, at variance with the SMV scheme where first the gauge-invariant part Ω_I is minimized through the disentanglement step, and only after the gauge-dependent part $\tilde{\Omega}$ is minimized through the usual MV scheme. The iterative minimization of the total spread Ω has been further developed by Damle, Levitt and Lin in Damle *et al.* (2019). They reformulated the Wannierization, which is a constrained non-linear optimization problem, as unconstrained optimization on matrix manifolds, where the SMV disentanglement procedure can be interpreted as a splitting method which represents an approximate solution.

We stress that even if minimization of the full the spread functional Ω guarantees the highest degree of overall localization, several of the methods discussed here can actually produce WFs such that a subset of them might individually be more localized than their maximally-localized counterparts. Along those lines, Fontana *et al.* developed the spread-balanced WFs (Fontana *et al.*, 2021), where they added a penalty term to the spread functional, proportional to the variance of the spread distribution among all WFs of the system. This scheme could be less prone to produce solutions with one or several poorly localized WFs, at the price of an increased total spread for the whole set. The addition of terms to the spread functional can also be used to pre-

serve some degree of locality in energy, such as in the case of mixed Wannier-Bloch functions (Giustino and Pasquarello, 2006) and dually localized Wannier functions (Mahler *et al.*, 2022). These approaches are based on a generalized spread functional (Gygi *et al.*, 2003) designed to carry both spatial localization (Wannier character) and limited spectral broadening (Bloch character), by minimizing a functional that contains not only a spatial variance (as for MLWF) but also an energy variance.

Once a choice for the functional to be minimized is made (the total spread as in the MLWF scheme, or any other choice), there is still a lot of flexibility on the choice of algorithm to perform the minimization. First, one needs to define a starting guess for the unitary matrices $U_{\mathbf{k}}$, which is customarily obtained by specifying a set of localized projection functions. While for composite bands a set of randomly centered spherically-symmetric Gaussian orbitals might work, in general more sophisticated choices are required. Typically, atomic orbitals are used as projection functions, such as s, p, d -orbitals as well as hybrid orbitals (e.g., sp^3), which are often centered either on atoms or along bond directions. As discussed in Sec. II.C.2, the choice of the right atomic orbitals is typically based on chemical intuition and can be partly informed by inspecting the projected density of states in the energy region of interest. Still, the choice of the right projection orbitals—i.e., those providing a good starting point for a successful minimization of the target functional—can often be a non-trivial task, especially in the context of automated high-throughput materials screening and, more generally, when one wants to study a novel material never investigated before (especially in case of unfamiliar orbital composition). Hence, in the last decade substantial efforts have been targeted at developing automated algorithms removing the need for users to define appropriate initial projections. A first approach in this direction is the optimized projection functions (OPFs) method (Mustafa *et al.*, 2015) for composite bands. In OPF, a larger set of functions that overspan the space of MLWFs is built and used as a starting point. While in plane-wave codes the OPF approach (Mustafa *et al.*, 2015) still needs the user to provide a list of initial projections, for instance atomic-like local orbitals (LOs), *ab initio* codes operating with localized (or mixed plane-wave/localized) basis sets can leverage the built-in localized orbitals. For instance, the full-potential linearized augmented-plane wave (LAPW) method can be extended by adding the so-called LOs, which are atomic-like, very localized, and can be employed in the construction of WFs. Tillack *et al.* (2020) combined the SMV disentanglement with the OPF method (Mustafa *et al.*, 2015) to construct initial guesses for MLWFs from a set of LOs in an automated way. Finally, another set of parameters that require tuning in the standard SMV disentanglement scheme are the inner and outer energy windows. Gresch *et al.* (2018) targeted the removal of the need for

manual input by focusing on the automated optimization of both windows.

On the algorithmic side, in the quest for fully automating the generation of localized WFs, various general and practical approaches have been recently proposed, targeting the construction of well-localized WFs using algorithms that are often non-iterative; this not only makes them more automatable, but also provides an excellent starting point for a final Wannierization, if required: the SCDM approach (Damle and Lin, 2018; Damle *et al.*, 2015, 2017), the continuous Bloch gauges (Cancès *et al.*, 2017; Gontier *et al.*, 2019), and the projectability disentanglement and manifold remixing approaches (Qiao *et al.*, 2023a,b).

SCDM is based on QR factorization with column pivoting (QRCP) of the reduced single-particle density matrix. The approach can be used either to produce well-localized WFs without performing an iterative minimization, or it can be considered a linear-algebra method to identify a good starting point for a MLWF procedure. The SCDM method is implemented (Vitale *et al.*, 2020) in the interface code to Quantum ESPRESSO, with an algorithm for the QRCP factorization that works on a smaller matrix instead of the full density matrix (Damle and Lin, 2018; Damle *et al.*, 2015, 2017). For a set of composite bands, SCDM is parameter-free. A comprehensive study on 81 insulators (Vitale *et al.*, 2020) has shown how the MLWF procedure applied to SCDM initial projections (SCDM+MLWF) improves the interpolation accuracy and localization of the resulting WFs, although SCDM-only WFs perform already remarkably well, both in terms of accuracy of band structure interpolation and in terms of localization. In the case of entangled bands, SCDM requires to specify an energy-window function, its center and width, as well as the number of WFs to consider. Vitale *et al.* (2020) introduced a recipe to automatically select those parameters (more details in Sec. III.J), which was tested on 200 bulk materials. Unlike in the case of isolated groups of bands, for entangled bands the SCDM+MLWF method greatly improved the localization of the WFs with respect to SCDM-only. Notably, however, the reduced spread induced by the MV procedure might result in lower band-interpolation accuracy. Although the SCDM projections could be used together with the SMV disentanglement scheme, this more complex procedure does not provide systematic gain in accuracy, such that a SCDM-only or SCDM+MLWF (with MV minimization for $\tilde{\Omega}$) is recommended (Vitale *et al.*, 2020). In addition, we note that SCDM requires real-space wavefunctions as input, and therefore has a higher computational and memory cost with respect to other methods discussed later in this section, which are implemented in reciprocal space.

Another non-iterative approach for composite bands is based on ensuring a continuous Bloch gauge over the entire BZ (Cancès *et al.*, 2017; Gontier *et al.*, 2019), re-

sulting in good localization properties of the WFs and not requiring any chemical intuition for their construction. The main idea is that one can construct a sequence of gauge matrices that are not only continuous across the BZ but also satisfy its periodicity at the BZ edges. This is achieved by first adopting parallel transport for the gauge matrix, starting from a chosen k -point (usually the Γ point) and propagating along a line (e.g. the k_x line). This allows to fix the periodicity at the two end points of the line, while preserving the continuity. Then, for each k -point on the line, parallel transport is applied to each of the gauge matrices along an orthogonal direction (e.g. in the k_y direction), and all gauge matrices are fixed again at the endpoints, to ensure periodicity of the 2D plane. Finally, for each k -point on the 2D plane, one can apply a similar procedure to construct gauge matrices along the third direction, therefore obtaining a global continuous gauge across the full BZ. Often, the resulting gauge is continuous but not smooth enough: a subsequent conventional MV iterative minimization can improve the localization and reach the MLWF gauge. Such algorithm is able to construct MLWFs for difficult cases such as $\mathbb{Z}_2$ topological insulators (Gontier *et al.*, 2019).

Lastly, a very robust approach has emerged in the form of projectability disentanglement (Qiao *et al.*, 2023b), where the inner and outer energy windows are replaced by projectability thresholds. For each state $|u_{n\mathbf{k}}\rangle$, a projectability (Agapito *et al.*, 2013; Vitale *et al.*, 2020) onto localized atomic orbitals (typically, those coming from the pseudopotentials) is calculated. Then, states that have very high projectability are retained identically (exactly as done for states inside the inner frozen window in the SMV method); states that have low projectability are discarded altogether (since they do not provide useful contributions to MLWFs); and states that span the intermediate projectability values are treated with the standard SMV disentanglement. This approach leads naturally and robustly to atomic-like projectability-disentangled Wannier functions (PDWFs), spanning both occupied and unoccupied states corresponding to Bloch sums of bonding/anti-bonding combinations of atomic orbitals.

These PDWFs can in turn be remixed into linear combinations that aim to describe target submanifolds; e.g., the valence states only, the conduction states only, or certain groups of bands that are separated in energy from the rest. This may be beneficial for finding optimal target states for beyond-DFT methods (see Sec. III.H).

This remixing is particularly valuable for Koopmans functionals (see Sec. III.H.2), which require separate sets of MLWFs for the valence and conduction manifolds, or for transport calculations. For this purpose, the manifold-remixed Wannier functions (MRWFs) (Qiao *et al.*, 2023a) are obtained by starting from the PDWFs spanning the whole manifold (valence plus conduction), which is then split by rotating the gauge matrices into

a block-diagonal structure across all the k -points, while simultaneously maintaining the gauge smoothness for each block. This is achieved by a combination of automated Wannierization of the whole manifold, diagonalization of the Wannier Hamiltonian, parallel transport, and maximal localization. The automated Wannierization of the whole manifold can be obtained using the PDWF method; the Hamiltonian diagonalization splits the manifold into desired submanifolds (e.g., two for valence and conduction, respectively); the parallel transport fixes the gauge randomness to construct two sets of localized WFs; the final maximal localization smoothens the gauge, leading to subsets of MLWFs for the respective submanifolds. Qiao *et al.* (2023a) demonstrates that, when combined with PDWFs, the MRWF method can be fully automated, and can also be extended to other types of band manifolds gapped in energy, such as the single top valence band of MoS₂, or the $3d$ and t_{2g}/e_g submanifolds of SrVO₃. For high-throughput results of PDWF and MRWF, see Sec. III.J.

III. The Wannier ecosystem: theory and software packages

A. Development of widely-available Wannier engines

The MV and SMV methods described in Marzari and Vanderbilt (1997) and Souza *et al.* (2001) were originally implemented in Fortran 77. The code would compute the overlaps in Eq. (8) and the projection of the periodic part of the Bloch orbitals onto trial localized states, by reading the former evaluated on a regular k -point grid by a DFT code—originally by an early version of CASTEP (Clark *et al.*, 2005; Marzari *et al.*, 1997). To provide a more general model, driven by the need to interface with a DFT code based on the LAPW method (Posternak *et al.*, 2002), the choice was made to keep the calculation of all the scalar products involving Bloch orbitals needed by the Wannier code within the electronic-structure code of choice, typically as a postprocessing step, and to establish well-defined protocols to exchange this information writing/reading files to/from disk; the format of those files was defined in the accompanying documentation, and the Wannier77 code was released under a GPL v2 license in March 2004.

In 2005, two of the current authors (AAM and JRY), then working in the groups of NM and IS respectively, rewrote the routines using modern modular Fortran, relying on their experience of software development gained from working on the ONETEP (Prentice *et al.*, 2020) and CASTEP (Clark *et al.*, 2005) DFT programs. The resulting program, Wannier90 (Mostofi *et al.*, 2008), was released under a GPL license in April 2006. Following the early layout in the Wannier77 code, Wannier90 was designed to be easily interfaced to any electronic structure code, irrespective of its underlying basis set. The first release

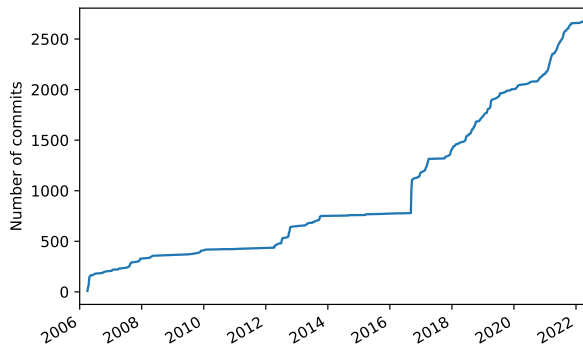


FIG. 3 Total number of commits over time for the **Wannier90** repository (CVS until August 2016, then transferred to Git and hosted on GitHub). Note the significant increase during and right after the community developer workshop in September 2016.

of **Wannier90** came with extensive documentation, tutorials, and two validation tests. Development used CVS as a version control system. GP joined the development effort in 2012, and a new parallel post-processing code (**postw90**) was developed and released in **Wannier90** version 2 (Mostofi *et al.*, 2014) in October 2013.

While the development of **Wannier90** as an open-source interoperable code was innovative in 2006, by 2016 it was clear that the development tools being used did not make use of what was then considered best practice. For example, having only a few developers with access to the main repository presented a barrier to adding new functionality to the program. A decision was made to move to a community development model, and the **Wannier90** repository was migrated to GitHub with the adoption of a “fork and pull request” approach. This new model was launched with a community developer workshop held in San Sebastián in September of 2016. This event is clearly recognizable in Fig. 3, which shows the number of commits to the code repository over time, with a large number of commits contributed during (or right after) the 2016 event. Moreover, once the code made its transition to a community development model, the rate of commits significantly increased, as well as the number of individual contributors (over 35 different people contributed code, tests or documentation via commits by the end of 2022). Essential to this change was the development of an extensive suite of tests, which runs automatically to validate every pull request. In 2019, **Wannier90** version 3 was released, including all community contributions to the code (Pizzi *et al.*, 2020).

In addition to **Wannier90**, there are a few electronic structure software packages that nowadays implement internal functionality for computing WFs in periodic systems. These include **GPAW** (Enkovaara

et al., 2010), **QBOX** (Gygi, 2008), **CP2K** (Kühne *et al.*, 2020), **exciting** (Tillack *et al.*, 2020), **OpenMX** (Weng *et al.*, 2009), **RESPACK** (Nakamura *et al.*, 2021) and **CRYSTAL** (Zicovich-Wilson *et al.*, 2001). However, in this review we want to use the term “Wannier engines” to describe software packages that generate WFs *and* are designed to be used interoperably with other software packages: e.g., with the electronic structure codes for solving the electronic ground state, used as input to a Wannier engine, and with post-processing codes that use the WFs from a Wannier engine to calculate advanced electronic properties. **Wannier90** is clearly an example of a Wannier engine, but it is not the only readily-available one—another notable example is the **Atomic Simulation Environment (ASE)** (Larsen *et al.*, 2017), which also implements routines based on the minimization of the quadratic spread of the WFs, but uses a different approach (Fontana *et al.*, 2021; Thygesen *et al.*, 2005) to that described above (see also Sec. II.D). In addition, the recent **Wannier.jl** package (Qiao *et al.*, 2023d) implements several Wannierization algorithms using manifold optimization techniques, and brings the methodology of WFs to the Julia (Bezanson *et al.*, 2017) community. In this review, we focus on those codes that interface to the **Wannier90** code; nevertheless, we note that often the other Wannier engines have adopted the same file formats first defined by **Wannier77/Wannier90** (see discussion in Sec. III.B), thus being fully compatible with the ecosystem.

B. The concept of a Wannier-functions software ecosystem

Complex software can adopt a variety of architecture design approaches, often differing substantially in the level of modularity (or lack thereof) of their components. Historically, most computer programs started as monolithic applications: self-contained and independent codes made of tightly coupled functions. This is a natural choice when writing new software from scratch, and it reduces the installation burden for users, who do not need to deal with the management of many dependencies. Over time, however, features and post-processing tools tend to get added, making the code-base very large and complex. This results in serious challenges for development and maintenance, which become critical when the code needs to be adapted and optimized for newer hardware architectures. Furthermore, this leads to reimplementation of common routines in each code, which could instead be written and optimized only once, and then used as a library. The library approach is already common in the electronic-structure community for linear-algebra and diagonalization routines, where the code calls functions via standard interfaces defined by the BLAS (Blackford *et al.*, 2002) and LAPACK (Anderson *et al.*, 1999) libraries,

and the executables are linked to performance-optimized versions on high-performance computing (HPC) clusters. While a similar approach is often used for other low-level routines, such as fast-Fourier-transform (FFT) computation (Frigo and Johnson, 2005) or to support file formats such as netCDF (Rew and Davis, 1990) or HDF5 (The HDF Group, 2023), it was until recently far less common for higher-level materials-science-oriented routines.

To address the challenges of monolithic codes, many electronic structure codes are being redesigned or rewritten using a more modular approach, where core modules are—when possible—generalized and separated into a library of reusable routines, then called by higher-level functions to execute complex tasks. Some of these codes have evolved into distributions, i.e., a set of relatively independent but interoperable executables reusing common core routines. However, even with this approach, the different modules can often operate only within the distribution, and the development of all modules needs to be constantly in sync.

Ultimate interoperability is obtained when code (such as core routines or full functionality) is reused by different independent software distributions, maintained by non-overlapping developer groups. A crucial challenge to enable such a level of interoperability is to design a clear application programming interface (API) defining which data needs to be transferred between codes, and in which format. This requires discussions and coordination, which can be catalyzed via targeted coordination efforts; an example worth mentioning is the CECAM Electronic Structure Library project (CECAM-ESL, 2023). At an even higher level, one can address code interoperability by defining common interfaces (e.g., input/output schemes) for workflows computing a quantity of interest, independent of the underlying simulation code, such as the common workflow interface of Huber *et al.* (2021) to perform crystal-structure relaxation and to compute equations of state. The workflow only requires as input, in a common format, the crystal structure and a few basic input parameters, and is then interfaced with 11 different DFT codes to run the actual simulations. Such universal interfaces make workflows accessible to a broader audience and codes fully interoperable, allowing researchers to switch between them without the need to learn from scratch the details of each one. In addition, they can be seamlessly applied to perform cross-code verification studies (Bosoni *et al.*, 2023).

When codes have to exchange data, the interfaces between them can be actual code APIs (e.g., in C or Fortran), where the library is directly compiled and linked with the main code, but also simply files in a well-documented format, written by the first application and read by the second one (see also later discussion of this approach in Sec. III.I). The actual choice depends on the interdependency between the algorithmic steps and on performance considerations. The use of files is typically

favored when the corresponding simulation workflows imply a sequential execution of codes rather than interconnected loops between them, when the exchanged data is small (up to few GB), and when the individual steps are computationally demanding, so that I/O overhead is only a small fraction of the total execution time. (In a few advanced cases, other interfaces such as network sockets have been used to keep the applications decoupled while still reducing the I/O overhead for simulations that are not very computationally demanding (Kapil *et al.*, 2019).) In addition, by writing intermediate results to files, the steps do not need to be combined in the same run, but can be executed at differed points in time (e.g., days later) or by different researchers.

In this context, WFs represent a remarkable and elegant method to decouple the *ab initio* simulation of the electronic structure from the calculation of the physical properties. This is possible thanks to two core aspects of WFs. First, WFs are independent of the basis set used in the first-principles electronic structure code: the MLWF algorithm requires the sole knowledge of a handful of vectors and matrices, such as the overlap matrices on a coarse grid of k points. Wavefunctions, typically stored in very large files, are not required during Wannierization, but only used optionally in few post-processing steps, e.g., when representing the WFs on a real-space grid. Second, many physical quantities can be obtained efficiently once a WF basis is constructed, only with the knowledge of relevant operators represented as small matrices directly in the Wannier basis, such as the Hamiltonian or the position operator. Indeed, while extended basis sets such as plane waves are particularly convenient to obtain charge densities and wavefunctions of periodic systems, reciprocal space integrals can be more efficiently calculated using a Fourier-interpolated basis set, originating from a compact maximally-localized representation in real space. From a computer-science perspective, we can say that these two aspects of the Wannierization process make it an effective data-compression encoding, avoiding the need to transfer large wavefunctions between the *ab initio* codes and the property calculators, while retaining an equivalent level of accuracy.

Thanks to the first aspect, i.e., basis-set independence, the Wannier code (Marzari and Vanderbilt, 1997) evolved from being a standalone code focused on the minimization procedure to one with a well-defined format for the input data (overlap and projection matrices), which also defined and documented the corresponding files (e.g., for the overlap matrices in `.mmn` format and the projection matrices in `.amn` format). The calculation of the latter was delegated to specific interfaces implemented within the corresponding first-principles packages (Posternak *et al.*, 2002). This design persisted in the Wannier90 code (Mostofi *et al.*, 2008; Pizzi *et al.*, 2020) and as a result the Wannier90 engine can now be interfaced with virtually any electronic-structure code as already dis-

cussed in Sec. III.A, with interfaces currently available for many widespread codes, including ABINIT (Gonze *et al.*, 2020), BigDFT (Ratcliff *et al.*, 2020), Elk (Elk, 2023), FLEUR (Wortmann *et al.*, 2023), GPAW (Enkovaara *et al.*, 2010), Octopus (Tancogne-Dejean *et al.*, 2020), OpenMX (Ozaki and Kino, 2005), pySCF (Sun *et al.*, 2018, 2020), Quantum ESPRESSO (Giannozzi *et al.*, 2017, 2009), SIESTA (Soler *et al.*, 2002), VASP (Kresse and Joubert, 1999) and WIEN2k (Blaha *et al.*, 2020).

Because of the second aspect, i.e., the possibility of efficiently obtaining many physical quantities in the Wannier basis, Wannier90 started to include a large number of efficient post-processing utilities for materials properties, ranging from simple band-structure interpolation to more complex properties such as the ordinary and anomalous Hall conductivities, Seebeck coefficients, orbital magnetization, and many more (Pizzi *et al.*, 2020). However, in the last decade the community has spontaneously moved towards a decentralized software ecosystem (as opposed to a centralized, albeit modular, Wannier distribution), where different packages interact through APIs and a common data format. The decentralized model was again facilitated by a clear and documented interface to generate data as input to the next steps (e.g., the `_tb.dat` file containing the full TB model: WF centers, on-site energies, and hopping energies). The community has been rapidly growing, and several independent packages exploiting MLWFs exist nowadays, targeting diverse properties such as TB models (Sec. III.C), ballistic transport (Sec. III.D), Berry-phase related properties (Sec. III.E), topological invariants (Sec. III.F), electron-phonon coupling (Sec. III.G), beyond-DFT methods (Sec. III.H), high-throughput calculations (Sec. III.J), and more.

This review article formalizes the existence of such a community of symbiotic packages, forming a research and software ecosystem built upon the concept of MLWFs. We illustrate this schematically in Fig. 2.

C. Wannier interpolation and tight-binding models

A very common application of WFs is to evaluate various k -space quantities and BZ integrals by “Wannier interpolation”. This name has come to refer to a type of Slater–Koster interpolation where the needed on-site and hopping integrals are calculated explicitly in the WF basis (Calzolari *et al.*, 2004; Lee *et al.*, 2005; Souza *et al.*, 2001; Yates *et al.*, 2007), as opposed to being treated as fitting parameters as done in empirical TB theory. Here we review the basic procedure as it applies to energy bands and other simple quantities, leaving more sophisticated applications to later sections. Before proceeding, let us mention that the Wannier interpolation scheme has been adapted to work with non-orthogonal localized orbitals instead of (orthogonal) WFs (Buongiorno Nardelli

et al., 2018; Jin *et al.*, 2021; Lee *et al.*, 2018; Wang *et al.*, 2019a).

1. Band interpolation

To interpolate the band structure, one needs the matrix elements of the KS Hamiltonian in the WF basis,

$$H_{ij}^W(\mathbf{R}) = \langle 0i | \hat{H} | \mathbf{R}j \rangle; \quad (11)$$

here $H_{ii}(\mathbf{0})$ are on-site energies, and the remaining matrix elements are hoppings. One way to evaluate these matrix elements is to start from Eq. (5a) for the WFs in terms of the KS Bloch eigenstates on the *ab initio* k -grid. Inserting that expression in Eq. (11) gives

$$H_{ij}^W(\mathbf{R}) = \frac{1}{N} \sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{R}} \sum_{n=1}^{J_{\mathbf{k}}} V_{\mathbf{k},ni}^* \varepsilon_{n\mathbf{k}} V_{\mathbf{k},nj}. \quad (12)$$

This procedure is particularly convenient in the framework of the MV and SMV Wannierization schemes, which are formulated as post-processing steps after a conventional *ab initio* calculation is carried out on a uniform $\{\mathbf{k}\}$ grid; Eq. (12) only involves the $V_{\mathbf{k}}$ matrices generated by the Wannier engine starting from the *ab initio* overlap matrices and energy eigenvalues (see Sec. II.A), and the energy eigenvalues themselves. An alternative to Eq. (12) would be to express the WFs in a real-space basis, e.g., localized orbitals or a grid, and then evaluate Eq. (11) directly on that basis.

In view of the localized character of the WFs, $|H_{ij}^W(\mathbf{R})|$ is expected to become negligibly small when the distance $|\mathbf{R} + \boldsymbol{\tau}_j - \boldsymbol{\tau}_i|$ between the centers of the two WFs becomes sufficiently large (here, $\boldsymbol{\tau}_j = \langle 0j | \hat{\mathbf{r}} | 0j \rangle$). However, due to the finite size N of the *ab initio* grid, the WFs obtained from Eq. (5a) are actually periodic over a real-space supercell of volume NV_{cell} ; accordingly, the matrix elements given by Eq. (12) are also supercell-periodic: $H_{ij}^W(\mathbf{R} + \mathbf{T}) = H_{ij}^W(\mathbf{R})$, for any supercell lattice vector $\mathbf{T}$. To minimize spurious effects associated with this artificial periodicity, the hopping matrix should be truncated by setting $H_{ij}^W(\mathbf{R}) = 0$ whenever the vector $\mathbf{R} + \mathbf{T} + \boldsymbol{\tau}_j$ lies outside the Wigner–Seitz (WS) supercell centered at the origin. Provided that this supercell is sufficiently large to ensure negligible overlap between a WF and its periodic images, the truncation error will be insignificant. This means that in practice one can achieve well-converged numerical results with a relatively coarse *ab initio* grid. Note, however, that the matrix elements do not decay exactly to zero for finite-size WS supercells. Therefore, when multiple $\mathbf{R}$ vectors lie on the boundary of the WS supercell and are connected by a supercell vector $\mathbf{T}$, it is better to consider all these equivalent vectors with appropriate weights, rather than picking only one of them, which would introduce spurious symmetry breaking in the Hamiltonian. The details of this approach and its

implementation in Wannier90 are discussed in Sec. 4.2 of Pizzi *et al.* (2020).

Once the on-site energies and hoppings have been tabulated, the Hamiltonian matrix is interpolated onto an arbitrary BZ point $\mathbf{k}'$ by performing an inverse Fourier transform,

$$H_{\mathbf{k}',ij}^W = \sum_{\mathbf{R}} \frac{1}{\mathcal{N}_{\mathbf{R},ij}} \sum_{l=1}^{\mathcal{N}_{\mathbf{R},ij}} e^{i\mathbf{k}' \cdot (\mathbf{R} + \mathbf{T}_{\mathbf{R},ij}^l)} H_{ij}^W(\mathbf{R}). \quad (13)$$

The summation runs over the lattice vectors $\mathbf{R}$ (which lie in the WS supercell centered at the origin, as discussed above) with $\mathcal{N}_{\mathbf{R}} > 1$ whenever $\mathbf{R} + \mathbf{T}_{\mathbf{R},ij}^l + \boldsymbol{\tau}_j$ falls on the boundary of the WS supercell centered at $\boldsymbol{\tau}_i$. To optimize the quality of the interpolation, for each combination of i, j , and $\mathbf{R}$ the supercell lattice vector $\mathbf{T}$ appearing in Eq. (13) is chosen as the one that minimizes $|\mathbf{R} + \mathbf{T} + \boldsymbol{\tau}_j - \boldsymbol{\tau}_i|$ (Pizzi *et al.*, 2020). Finally, the interpolated energy eigenvalues are obtained by diagonalizing the above matrix,

$$\left[\mathcal{U}_{\mathbf{k}'}^\dagger H_{\mathbf{k}'}^W \mathcal{U}_{\mathbf{k}'} \right]_{mn} = \delta_{mn} \epsilon_{n\mathbf{k}'}^H, \quad (14)$$

so that the column vectors of the unitary matrix $\mathcal{U}_{\mathbf{k}'}$ are eigenvectors of $H_{\mathbf{k}'}^W$.

Since the interpolation steps (13) and (14) only involve Fourier transforming and diagonalizing $J \times J$ matrices that are typically small, the overall procedure tends to be much less expensive than a direct DFT calculation at every interpolation point, especially when a dense interpolation grid $\{\mathbf{k}'\}$ is needed. The efficient evaluation of the Hamiltonian matrix and band derivatives (see below) allows to explore BZ integration beyond the standard equispaced scheme when fine features in k -space need to be resolved using adaptive integration methods (Assmann *et al.*, 2016; Kaye *et al.*, 2023).

The above interpolation scheme has been shown to accurately reproduce—within the frozen energy window—the energy eigenvalues obtained by a direct DFT calculation. As an example, we show in Fig. 4 a detail of the interpolated band structure of ferromagnetic body-centred cubic (BCC) Fe along the H– Γ line (Yates *et al.*, 2007). The vertical gray lines indicate points on the $\{\mathbf{k}\}$ mesh used for constructing the WFs. For comparison, we plot as plus symbols the *ab initio* dispersion around a weak spin-orbit-induced avoided crossing between two bands of opposite spin. It is apparent that the Wannier interpolation procedure succeeds in resolving details on a scale much smaller than the spacing between those points. In particular, the correct band connectivity is obtained, so that avoided crossings, no matter how weak, are not mistaken for actual crossings. This characteristic, which distinguishes Wannier interpolation from methods based on direct Fourier interpolation of the energy eigenvalues (Madsen *et al.*, 2018; Madsen and Singh, 2006),

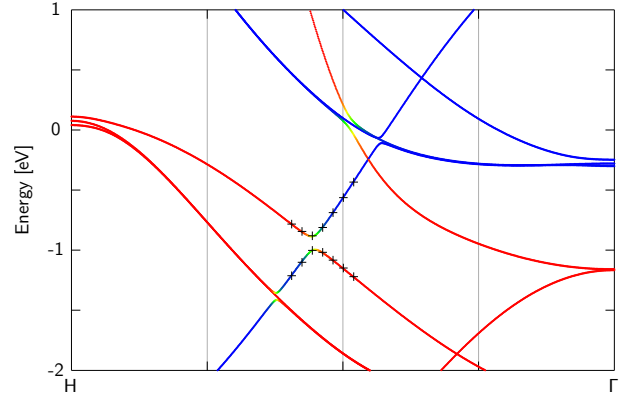


FIG. 4 Wannier-interpolated bands of BCC Fe along the H– Γ line. The bands are color-coded according to the value of the spin expectation value $\langle \psi_{n\mathbf{k}} | \hat{S}_z | \psi_{n\mathbf{k}} \rangle$: red for spin up, and blue for spin down. The energies are given in eV, and the Fermi level is at 0 eV. The vertical gray lines indicate k -points on the *ab initio* mesh used for constructing the WFs.

makes it a powerful tool for studying topological properties (Sec. III.F), and for evaluating BZ integrals involving quantities that change rapidly over small regions of k -space, such as the Berry curvature (Sec. III.E) and electron-phonon matrix elements (Sec. III.G).

Wannier interpolation works for the same reason that empirical TB does: the short range of the real-space Hamiltonian matrix (12) ensures that its Fourier transform (13) is a smooth function in reciprocal space. This can also be seen by writing the left-hand side of Eq. (13) as

$$H_{\mathbf{k}',ij}^W = \langle \psi_{i\mathbf{k}'}^W | \hat{H} | \psi_{j\mathbf{k}'}^W \rangle, \quad (15)$$

where

$$|\psi_{j\mathbf{k}'}^W\rangle = \sum_{\mathbf{R}} e^{i\mathbf{k}' \cdot \mathbf{R}} |\mathbf{R}_j\rangle \quad (16)$$

interpolates the smooth Bloch functions defined on the *ab initio* grid by Eq. (5b). We may also write the left-hand side of Eq. (14) as

$$H_{\mathbf{k}',mn}^H = \langle \psi_{m\mathbf{k}'}^H | \hat{H} | \psi_{n\mathbf{k}'}^H \rangle, \quad (17)$$

where

$$|\psi_{n\mathbf{k}'}^H\rangle = \sum_{j=1}^J |\psi_{j\mathbf{k}'}^W\rangle \mathcal{U}_{\mathbf{k}',jn} \quad (18)$$

describes a unitary transformation from the Wannier gauge W to the Hamiltonian gauge H. Inside the frozen energy window, the states $|\psi_{n\mathbf{k}'}^H\rangle$ interpolate—up to arbitrary phase factors—the *ab initio* eigenstates $|\psi_{n\mathbf{k}}\rangle$.

In summary, performing Fourier interpolation in the W gauge followed by a unitary transformation to the H gauge allows interpolating quantities—band energies,

Bloch eigenstates, and matrix elements thereof (see below)—that can vary rapidly in k -space, and even become non-analytic at degeneracies. This strategy retains the accuracy of a full-blown *ab initio* calculation, while benefiting from the efficiency of Slater–Koster interpolation.

2. Band derivatives and Boltzmann transport

The interpolation procedure outlined above can be adapted to evaluate band velocities, inverse effective-mass tensors, and higher k -derivatives of the energy eigenvalues (Yates *et al.*, 2007); as in the empirical TB method (Graf and Vogl, 1995), this is achieved without

relying on finite-difference methods, which become problematic in the vicinity of band crossings and weak avoided crossings, where the band ordering can change from one grid point to the next.

Band derivatives are needed, for instance, to evaluate transport coefficients such as the electrical conductivity σ , the Seebeck coefficient S , or the electronic contribution to the thermal conductivity K . Within the semi-classical Boltzmann transport equation (BTE) framework, one defines a scattering time $\tau_{n\mathbf{k}}$ for an electron on band n at wavevector $\mathbf{k}$ (incidentally, the contributions from electron–phonon scattering to $\tau_{n\mathbf{k}}$ can be efficiently computed exploiting Wannier functions, see Sec. III.G). Then, the expressions for the transport tensors are given by (Ziman, 1972):

$$\sigma_{ab}(\mu, T) = e^2 \int_{-\infty}^{+\infty} dE \left(-\frac{\partial f(E, \mu, T)}{\partial E} \right) \Sigma_{ab}(E), \quad (19)$$

$$[\sigma S]_{ab}(\mu, T) = \frac{e}{T} \int_{-\infty}^{+\infty} dE \left(-\frac{\partial f(E, \mu, T)}{\partial E} \right) (E - \mu) \Sigma_{ab}(E), \quad (20)$$

$$K_{ab}(\mu, T) = \frac{1}{T} \int_{-\infty}^{+\infty} dE \left(-\frac{\partial f(E, \mu, T)}{\partial E} \right) (E - \mu)^2 \Sigma_{ab}(E), \quad (21)$$

where μ is the chemical potential, T is the temperature, a and b are Cartesian indices, σS denotes the matrix product of the two tensors, $\partial f/\partial E$ is the derivative of the Fermi–Dirac distribution function with respect to the energy, and $\Sigma_{ab}(E)$ is the transport distribution function. The latter is defined by

$$\Sigma_{ab}(E) = \frac{1}{V_{\text{cell}}} \sum_{n\mathbf{k}} v_{n\mathbf{k}}^a v_{n\mathbf{k}}^b \tau_{n\mathbf{k}} \delta(E - E_{n\mathbf{k}}), \quad (22)$$

where the summation is over all bands n and over all the BZ, $E_{n\mathbf{k}}$ is the energy for band n at $\mathbf{k}$ and $v_{n\mathbf{k}}^a$ is the a component of the band velocity at $(n, \mathbf{k})$.

Obtaining converged quantities for Eqs.(19)–(21), therefore, requires to compute the band derivatives $\mathbf{v}_{n\mathbf{k}}$ while sampling the BZ over dense k -point grids (Madsen and Singh, 2006; Schulz *et al.*, 1992; Uehara and Tse, 2000), since the term $\partial f/\partial E$ is non-zero only in a narrow energy region (of typical size $k_B T$, where k_B is the Boltzmann constant) around the chemical potential μ . Wannier interpolation allows carrying out this task efficiently and accurately even when (avoided) crossings occur close to the Fermi level: band derivatives at a given k -point are obtained with an analytical expression, without resorting to finite-difference methods (Yates *et al.*, 2007). Moreover, computation on dense k -point grids is very efficient, as already discussed earlier for band interpolation. This WF-based Boltzmann-transport methodology

is implemented in Wannier90 and used to compute transport tensors in its BoltzWann module (Pizzi *et al.*, 2014), as well as in other codes (see, e.g., the electron-phonon section, Sec. III.G), and is also used for post-processing calculations in many-body theory (see Sec. III.H.1).

Furthermore, many transport coefficients (e.g., linear and non-linear (AHCs) (Sodemann and Fu, 2015; Yao *et al.*, 2004), anomalous Nernst thermoelectric conductivity (Xiao *et al.*, 2006), magnetoresistance (Gao *et al.*, 2017), and magnetochiral anisotropy (Yokouchi *et al.*, 2023)) depend on the Berry curvature and other quantum-geometric quantities (Gao, 2019; Vanderbilt, 2018; Xiao *et al.*, 2010). As they involve k -derivatives of the Bloch states themselves, such quantities cannot be obtained from the energy dispersions. Moreover, those quantities tend to become strongly enhanced when weak avoided crossings occur near the Fermi level; when that happens, very dense k -point grids must be employed to converge the calculation (Yao *et al.*, 2004). Compared to band interpolation, the interpolation of Berry-type quantities is more involved because it requires setting up matrix elements of the position operator $\hat{\mathbf{r}}$, which is non-local in reciprocal space (Blount, 1962). We defer a discussion of that case to Sec. III.E, and below we describe how to interpolate the matrix elements of a generic k -local operator $\hat{X}$.

3. Interpolation of a generic k -local operator

Replacing $\hat{H} \rightarrow \hat{X}$ in Eq. (11) and using Eq. (5) yields

$$X_{ij}^W(\mathbf{R}) = \frac{1}{N} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{R}} \sum_{m,n=1}^{J_{\mathbf{k}}} V_{\mathbf{k},mi}^* \langle \psi_{m\mathbf{k}} | \hat{X} | \psi_{n\mathbf{k}} \rangle V_{\mathbf{k},nj}, \quad (23)$$

which reduces to Eq. (12) for $\hat{X} = \hat{H}$. The considerations made earlier regarding the spatial decay and truncation of the $H^W(\mathbf{R})$ matrix apply equally well to $X^W(\mathbf{R})$. The Fourier-transform step is also analogous to Eq. (13),

$$X_{\mathbf{k}',ij}^W = \sum_{\mathbf{R}} e^{i\mathbf{k}' \cdot \mathbf{R}} X_{ij}^W(\mathbf{R}) = \langle \psi_{i\mathbf{k}'}^W | \hat{X} | \psi_{j\mathbf{k}'}^W \rangle, \quad (24)$$

and the final step is to apply to $X_{\mathbf{k}'}^W$ the same unitary transformation that was used in Eq. (14) to diagonalize $H_{\mathbf{k}'}^W$. Using Eq. (18), we find

$$X_{\mathbf{k}',mn}^H = \left(\mathcal{U}_{\mathbf{k}'}^\dagger X_{\mathbf{k}'}^W \mathcal{U}_{\mathbf{k}'} \right)_{mn} = \langle \psi_{m\mathbf{k}'}^H | \hat{X} | \psi_{n\mathbf{k}'}^H \rangle. \quad (25)$$

In particular, diagonal elements of $X_{\mathbf{k}'}^H$ give the expectation values of $\hat{X}$ in the interpolated states. With $\hat{X} = \hat{S}$, for example, one obtains their spin polarization, which is how the color-coding in Fig. 4 was generated.

Note that in addition to the overlap matrices and energy eigenvalues, interpolating a generic operator $\hat{X} \neq \hat{H}$ requires setting up its matrix elements $\langle \psi_{m\mathbf{k}} | \hat{X} | \psi_{n\mathbf{k}} \rangle$ on the *ab initio* grid; this should be done by the same interface code that computes the overlap matrices.

4. Wannier function perturbation theory

Several important materials properties can be calculated as the linear response of the system to an external perturbation $\hat{V}$. A common example is the calculation of phonon dispersions and electron-phonon coupling through density-functional perturbation theory (DFPT) (Baroni *et al.*, 2001), the latter case is discussed in Sec. III.G.

We follow Lihm and Park (2021), and write the Hamiltonian eigenstates of the perturbed system in terms of the ones of the unperturbed system plus the wavefunction perturbation,

$$|\psi_{n\mathbf{k}}\rangle = |\psi_{n\mathbf{k}}\rangle^{(0)} + \lambda |\psi_{n\mathbf{k}}\rangle^{(1)} + \mathcal{O}(\lambda^2), \quad (26)$$

where the wavefunction perturbation can be calculated with a sum over empty states,

$$|\psi_{n\mathbf{k}}\rangle^{(1)} = \sum_{n'\mathbf{k}'}' |\psi_{n'\mathbf{k}'}\rangle^{(0)} \frac{\langle \psi_{n'\mathbf{k}'} | \hat{V} | \psi_{n\mathbf{k}} \rangle^{(0)}}{\epsilon_{n\mathbf{k}} - \epsilon_{n'\mathbf{k}'}}. \quad (27)$$

The primed sum means that terms for which the denominator vanishes are excluded.

Alternatively, the wavefunction perturbation can be calculated without summing over high-energy states by solving the Sternheimer equation (Baroni *et al.*, 2001). Lihm and Park (Lihm and Park, 2021) have shown that such perturbation theory can be formulated in the Wannier representation, where the WFs of the perturbed system can be written as

$$|\mathbf{R}j\rangle = |\mathbf{R}j\rangle^{(0)} + \lambda |\mathbf{R}j\rangle^{(1)} + \mathcal{O}(\lambda^2). \quad (28)$$

The expression for $|\mathbf{R}j\rangle^{(1)}$, reported in Eqs. (8) and (9) of Lihm and Park (2021), consists of two terms: the first can be calculated with the Sternheimer equation to obtain the states $|\psi_{n\mathbf{k}}\rangle^{(1)}$, while the second contains matrix elements of $\hat{V}$ and the projector over the WFs of the unperturbed system. Crucially, both terms only require energies and matrix elements within the Wannier outer window introduced in Sec. II.A.2. Thus, the Wannier function perturbation can be calculated without making explicit use of the states outside that energy window.

For a monochromatic static perturbation with wavevector $\mathbf{q}$, the first-order wavefunction perturbation can be interpolated as (Lihm and Park, 2021)

$$|\psi_{n\mathbf{k}',\mathbf{q}}^H\rangle^{(1)} = \frac{1}{\sqrt{N}} \sum_{j,\mathbf{R}} e^{i\mathbf{k}' \cdot \mathbf{R}} |\mathbf{R}qj\rangle^{(1)} \mathcal{U}_{\mathbf{k}',jn} + \sum_{m=1}^{N_W'} |\psi_{m\mathbf{k}'+\mathbf{q}}^H\rangle^{(0)} \frac{\tilde{g}_{mn\mathbf{k}',\mathbf{q}}^H}{\epsilon_{n\mathbf{k}'}^{(0)} - \epsilon_{m\mathbf{k}'+\mathbf{q}}^{(0)}}, \quad (29)$$

where the WF perturbations are expanded as a sum of monochromatic perturbations

$$|\mathbf{R}j\rangle^{(1)} = \sum_{\mathbf{q}} |\mathbf{R}qj\rangle^{(1)} \quad (30)$$

while the superscript H marks the Hamiltonian gauge of the unperturbed (0) and perturbed (1) system, $\mathcal{U}_{\mathbf{k}'}$ is the unitary matrix that diagonalizes the unperturbed Hamiltonian according to Eq. (14), and $\tilde{g}_{mn\mathbf{k},\mathbf{q}}^H$ is obtained by performing the Fourier transform of the matrix elements of $\hat{V}$ between WFs of the unperturbed states, and then rotating to the Hamiltonian gauge using $\mathcal{U}_{\mathbf{k}'}$.

A key aspect is that Wannier function perturbations are localized in real space, so the perturbed Hamiltonian and its eigenstates can be efficiently interpolated by considering coarse k -point grids. This allows to efficiently interpolate various matrix elements involving the wavefunction perturbation, such as in the case of electron-phonon self-energies and Kubo formulas. Wannier function perturbation theory (WFPT) has been applied to describe temperature-dependent electronic band structures and indirect optical absorption, shift spin currents and spin Hall conductivities (Lihm and Park, 2021). WFPT will be made available in a future release of the EPW code (Lee *et al.*, 2023).

5. Porting Wannier Hamiltonians to TB codes

In TB theory, two phase conventions are commonly used to perform the Fourier transforms from real to reciprocal space (Vanderbilt, 2018): the one adopted in Eqs. (13), (16) and (24) (“original convention”), and the alternative one (“modified convention”) where the phase factors in those equations are modified as

$$e^{i\mathbf{k}' \cdot \mathbf{R}} \rightarrow e^{i\mathbf{k}' \cdot (\mathbf{R} + \boldsymbol{\tau}_j - \boldsymbol{\tau}_i)}. \quad (31)$$

Although the interpolated eigenvalues $\varepsilon_{n\mathbf{k}'}^H$ and matrix elements $X_{\mathbf{k}',mn}^H$ come out the same with both conventions (as they should), the modified convention is more natural for the purpose of evaluating quantities, such as Berry connections and curvatures, that are sensitive to the real-space embedding of the TB model via the position matrix elements $\langle 0i|\hat{\mathbf{r}}|\mathbf{R}j\rangle$ (see Sec. III.E). This has to do with the fact that with the original convention, TB eigenvectors (the column vectors of $\mathcal{U}_{\mathbf{k}'}$) behave like full Bloch eigenstates $|\psi_{n\mathbf{k}'}\rangle$, whereas with the modified convention they behave like their cell-periodic parts $|u_{n\mathbf{k}'}\rangle$ (Vanderbilt, 2018); and it is in terms of the latter that Berry-type quantities are most naturally expressed.

The modified phase convention is the one adopted in the TB codes `PythTB` (PythTB, 2023a) and `TBmodels` (TBmodels, 2023); both are able to import the Wannier Hamiltonian $H_{ij}^W(\mathbf{R})$, along with the Wannier centers $\{\boldsymbol{\tau}_j\}$, from the `seedname.tb.dat` file written by `Wannier90`. `PythTB` was originally written for pedagogical purposes, as part of a course on Berry phases in electronic structure theory that was later turned into a textbook (Vanderbilt, 2018). It is feature-rich but is not optimized for speed, as it was designed with TB toy models in mind (however, a high-performance implementation is available (Numba-PythTB, 2023)). Instead, `TBmodels` has fewer post-processing functionalities, but it delivers critical speed-up and improved scaling. Among other Wannier-TB codes, it is worth mentioning `Wannier Tools` (Wu *et al.*, 2018), which implements sparse Hamiltonians for large systems, band unfolding, and several other features related to Berry-type quantities (see Sec. III.E).

6. Wannier interpolation beyond density-functional theory

As discussed above, one of the most powerful and effective applications of WFs is the interpolation of band structures and other electronic-structure properties. While this is already very useful in the context of DFT calculations, it becomes even more compelling for beyond-DFT methods, such as hybrid functionals (Becke, 1988, 2014; Heyd *et al.*, 2003; Kümmel and Kronik, 2008; Lee *et al.*, 1988; Martin, 2020; Perdew *et al.*, 1996), many-body perturbation theory (MBPT) such as G_0W_0 (Golze *et al.*, 2019), and non-diagrammatic approaches such

as the Koopmans-compliant functionals (Borghi *et al.*, 2014; Colonna *et al.*, 2022, 2018, 2019; Dabo *et al.*, 2010; De Gennaro *et al.*, 2022; Elliott *et al.*, 2019; Nguyen *et al.*, 2018). In fact, in DFT the potential can always be recalculated from the sole knowledge of the ground-state electronic charge density; therefore, the corresponding KS Hamiltonian can be directly calculated at any arbitrary $\mathbf{k}$ -point. Instead, for most beyond-DFT methods this is no longer the case, and band structure calculations on a high-symmetry path cannot be performed as a series of independent diagonalizations. For hybrid functionals and GW, the eigenvalues at a given $\mathbf{k}$ -point requires the knowledge of the wavefunctions and eigenenergies on all $(\mathbf{k} + \mathbf{q})$ -points, where the $\mathbf{q}$ -points are defined on a uniform grid which has to be converged. In other words, reasonably dense sampling on high-symmetry paths can only be obtained with some form of interpolation.

While electronic structure codes typically offer general-purpose interpolation methods, often based on Fourier series (Koelling and Wood, 1986; Pickett *et al.*, 1988), WFs provide two concrete advantages. First, they are a physically motivated and optimal basis set, which exhibits exponential convergence and is guaranteed to deliver the exact result for a sufficiently dense $\mathbf{k}$ -point grid, so the accuracy can be systematically increased simply by considering denser grids. If MLWFs are chosen, the efficiency is maximal and rather coarse grids are often sufficient to faithfully reproduce the entire band structure. Second, once a WF basis is constructed, not only it yields interpolated eigenvalues (e.g., the band structure) but it also allows to represent the Hamiltonian and many other operators in a compact real space representation. Once the relevant operators in a WF basis are available, one gets access to the full spectrum of theories and software packages that are part of the Wannier ecosystem, capable of much more complex tasks than just band interpolation. Notably, once a Wannierization is performed with hybrid functionals or at the G_0W_0 level, all other Wannier-interpolated quantities can be obtained at the same level of theory with no extra effort or cost. Finally, thanks to all the recent work in advanced minimization techniques (Sec. II.D) and automation (Sec. III.J), the Wannier interpolation does not require much more human intervention than other standard methods such as smooth Fourier interpolation. In the following, we briefly outline the motivation and the corresponding procedure to deploy Wannier interpolation for two of the most popular excited-state approaches: hybrid functionals, and many-body perturbation theory at the G_0W_0 level.

a. Hybrid functionals. A very popular approach to improve the accuracy of *ab initio* band structures is to combine explicit density-dependent functionals with Hartree–Fock terms, which leads to orbital-dependent functionals called “hybrids” (Becke, 1988, 2014; Heyd

et al., 2003; Kümmel and Kronik, 2008; Lee *et al.*, 1988; Martin, 2020; Perdew *et al.*, 1996). The procedure to obtain WFs is similar to that in vanilla DFT, except that non-self-consistent field calculations cannot be performed. Hence, the self-consistent field calculation must already include all the empty states (if any) to be included in the Wannierization. In addition, ground-state calculations are typically performed on the irreducible Brillouin zone (IBZ) by exploiting crystalline symmetries, while **Wannier90** requires a uniform grid on the full Brillouin zone (FBZ). As performing the self-consistent calculation on the FBZ is certainly possible but rather inefficient, the typical procedure involves unfolding the ground-state orbitals and band structure from the IBZ to the FBZ as a post-processing step to be performed after the self-consistent calculation and before producing the overlap matrices and the other input required to obtain WFs. For example, in the **Quantum ESPRESSO** distribution (Giannozzi *et al.*, 2017, 2009) this is done through the `open_grid.x` code. Notably, WFs can be used to speed up the core hybrid-functionals calculations, as they allow reducing the number of exchange integrals to be computed (Carnimeo *et al.*, 2019; DiStasio *et al.*, 2014; E *et al.*, 2007; García-Cervera *et al.*, 2009; Mounjoy *et al.*, 2017; Wu *et al.*, 2009).

b. G_0W_0 . Most of what has been discussed for hybrid functionals also holds for MBPT calculations in the G_0W_0 approximation, with two important remarks. First, G_0W_0 is a one-shot approach in the quasiparticle (QP) approximation which is typically performed on top of a DFT calculation: so the orbitals remain at the KS level and only the energy eigenvalues are corrected, hence neglecting off-diagonal elements of the self-energy in the KS basis. Second, as only the energies are changed at the G_0W_0 level, the KS states might swap their band indices and not be ordered in energy anymore. A typical case where this might manifest clearly are topological insulator candidates (and systems with band inversions in general) such as monolayer TiNI (Marrazzo *et al.*, 2019), which is topological in DFT and trivial at the G_0W_0 level. The practicalities of obtaining G_0W_0 WFs and related quantities depend on whether the DFT and MBPT calculations are performed with the same distribution (e.g., **VASP** (Kaltak, 2015)) or with two separate codes (as for example with **Quantum ESPRESSO** (Giannozzi *et al.*, 2017, 2009) and **Yambo** (Sangalli *et al.*, 2019)). In general, G_0W_0 QP corrections $\Delta\epsilon_i^{\text{QP}} = \epsilon_i^{G_0W_0} - \epsilon_i^{\text{DFT}}$ need to be computed on a uniform grid in the FBZ, which can be made efficient by unfolding from the IBZ to the FBZ (in **Yambo**, this operation is carried out by the post-processing utility `ypp`). While orbitals remain at the DFT-KS level, **Wannier90** requires the states to be ordered with ascending energy and, in addition, some input matrices (e.g., the `uHu`) need to be updated with

QP corrections. If this is not performed by the *ab initio* engine, it can be taken care by the **Wannier90** utility `gw2wannier90.py` if the DFT eigenvalues and QP corrections are provided in the standard format `seedname.eig`. After this step, the Wannierization can proceed as usual, and it is available in both **Wannier90** (Pizzi *et al.*, 2020) and **WanT** (Ferretti *et al.*, 2012). Note that in G_0W_0 , QP corrections need to be computed on a subset of the k -point grid required to calculate the self-energy: this can be exploited to speedup the calculation, especially for 2D materials as discussed in Sangalli *et al.* (2019), because Wannierization typically requires relatively coarse k -point grids. Finally, we briefly touch on beyond- G_0W_0 development of interest from a WF perspective. While Aguilera *et al.* have shown that off-diagonal components of the self-energy—which are not included in standard perturbative approaches—are very relevant in case of band inversions (Aguilera *et al.*, 2013) (such as for topological insulators), Hamann and Vanderbilt have found (Hamann and Vanderbilt, 2009) that, in general, differences between MLWFs obtained with local-density approximation (LDA) and self-consistent GW (QSGW) are minimal.

c. Bethe–Salpeter equation In order to address neutral excitations, as opposed to the charged excitations of GW, one needs to describe the bound state of an excited electron with the hole that has been created. This is accomplished either using time-dependent density-functional theory or Green’s function methods (Onida *et al.*, 2002); in the latter case, the Bethe–Salpeter equation (BSE) is solved on top of the GW solutions. Within the Tamm–Dancoff approximation (Dancoff, 1950; Hirata and Head-Gordon, 1999; Onida *et al.*, 2002), the BSE can be recast into an effective two-particle eigenvalue problem (Onida *et al.*, 2002; Rohlfing and Louie, 2000), which in the electron-hole (e-h) basis reads

$$\sum_{v'c'} [D_{vc,v'c'} + 2K_{vc,v'c'}^x - K_{vc,v'c'}^d] A_{v'c'}^\lambda = E_\lambda A_{vc}^\lambda, \quad (32)$$

where the v, v' and c, c' indices run over valence and conduction states, E_λ are the neutral excitation energies (the poles of the density-density response function), and A_{vc}^λ are the coefficients of the excitonic wavefunctions (that are related to oscillator strengths) in the e-h basis. The effective 2-particle Hamiltonian is made by i) a diagonal term $D_{vc,v'c'} = (\epsilon_c^{\text{QP}} - \epsilon_v^{\text{QP}}) \delta_{vv'} \delta_{cc'}$ representing the “bare” e-h transitions (i.e., without accounting for the electron-hole interaction) from the QP theory, ii) an exchange-like term

$$K_{vc,v'c'}^x = \int d\mathbf{r} d\mathbf{r}' \phi_c^*(\mathbf{r}) \phi_v(\mathbf{r}) |\mathbf{r} - \mathbf{r}'|^{-1} \phi_{v'}^*(\mathbf{r}') \phi_{c'}(\mathbf{r}'), \quad (33)$$

and iii) a direct screened Coulomb term

$$K_{vc,v'c'}^d = \int d\mathbf{r} d\mathbf{r}' \phi_c^*(\mathbf{r}) \phi_{c'}(\mathbf{r}) W(\mathbf{r}, \mathbf{r}') \phi_{v'}^*(\mathbf{r}') \phi_v(\mathbf{r}') \quad (34)$$

responsible for an effective attractive interaction between the electron and the hole. Here $\{\phi_i\}$ and $\{\varepsilon_i^{\text{QP}}\}$ are the QP wavefunctions and QP energies, respectively. The solution of Eq. (32) in the e-h basis would require the explicit computation of a significant number of empty states, which becomes in turn a critical convergence parameter. This can be conveniently avoided (Giustino *et al.*, 2010; Marsili *et al.*, 2017; Rocca *et al.*, 2010; Umari *et al.*, 2011) resorting to well established techniques from DFPT (Baroni *et al.*, 2001) by introducing i) the projector over the conduction manifold $\hat{Q} = \mathbb{1} - \hat{P} = \mathbb{1} - \sum_v |\phi_v\rangle\langle\phi_v|$, and ii) a set of auxiliary functions $\xi_v(\mathbf{r}) = \sum_c A_{cv} \phi_c(\mathbf{r})$, usually called a *batch* representation (Rocca *et al.*, 2008; Walker *et al.*, 2006). These are N_v (where N_v is the number of occupied states) auxiliary functions that live in the unperturbed empty-states manifold and provide an equivalent but more compact representation of the excitonic wavefunction $\Theta(\mathbf{r}, \mathbf{r}') = \sum_{vc} A_{cv} \phi_v^*(\mathbf{r}) \phi_c(\mathbf{r}') = \sum_v \phi_v(\mathbf{r}) \xi_v(\mathbf{r}')$. In this representation, the effective 2-particle Hamiltonian is completely specified by its action on the components of the batch (Rocca *et al.*, 2010):

$$\sum_{v'} D_{vv'} |\xi_{v'}\rangle = \sum_{v'} (\hat{H}^{\text{QP}} - \varepsilon_{v'} \mathbb{1}) \delta_{vv'} |\xi_{v'}\rangle \quad (35)$$

$$\sum_{v'} K_{vv'}^x |\xi_{v'}\rangle = \sum_{v'} \hat{Q} \left(\int \frac{1}{|\mathbf{r} - \mathbf{r}'|} \phi_{v'}^*(\mathbf{r}') \xi_{v'}(\mathbf{r}') d\mathbf{r}' \right) |\phi_v\rangle \quad (36)$$

$$\sum_{v'} K_{vv'}^d |\xi_{v'}\rangle = \sum_{v'} \hat{Q} \left(\int W(\mathbf{r}, \mathbf{r}') \phi_{v'}^*(\mathbf{r}') \phi_{v'}(\mathbf{r}') d\mathbf{r}' \right) |\xi_{v'}\rangle. \quad (37)$$

The advantage of this formulation is that there is no explicit reference to the empty states (hidden inside the projector $\hat{Q}$ and the batch representation), and only the N_v auxiliary functions $\{|\xi_v\rangle\}$ need to be determined solving the BSE in the batch representation.

A further computational speed-up (and an improvement on the overall scaling (Marsili *et al.*, 2017)) can be achieved by moving from the KS orbitals to MLWFs and by exploiting their localization to greatly reduce the number of operations needed to evaluate the action of the BSE Hamiltonian on a trial state $|\xi_v\rangle$. This is particularly relevant for the direct term (Eq. (37)), which represents the real bottleneck of the calculations. In the Wannier representation, this becomes

$$\sum_{v'} K_{vv'}^d |\tilde{\xi}_{v'}\rangle = \hat{Q} \left(\sum_{v'} \int W(\mathbf{r}, \mathbf{r}') \omega_{v'}^*(\mathbf{r}') \omega_{v'}(\mathbf{r}') d\mathbf{r}' \right) |\tilde{\xi}_{v'}\rangle, \quad (38)$$

where $\{\omega_v(\mathbf{r})\}$ are the MLWFs and $\tilde{\xi}_v(\mathbf{r})$ is the batch component in the Wannier representation (simply obtained by rotating the original $\xi_v(\mathbf{r})$ with the unitary matrix rotation that transforms the manifold from the canonical to the MLWF representation). Exploiting locality, one can define a threshold for which a given pair of MLWFs overlap. By excluding non-overlapping pairs of MLWFs from the summation in Eq. (38), it becomes possible to lower the scaling of the evaluation of the action of the direct term on trial states from $\mathcal{O}(N^4)$ to $\mathcal{O}(N^3)$ (Marsili *et al.*, 2017), an approach that has been established and applied in the community (Elliott *et al.*, 2019; Marsili *et al.*, 2017; Umari *et al.*, 2011). Tight-binding and phenomenological models based on localized representation have also recently appeared (Dias *et al.*, 2023; Uría-Álvarez *et al.*, 2023).

D. Ballistic transport and nanostructures

MLWFs can be used to build the electronic structure of large nanostructures (Lee *et al.*, 2005; Lihm and Park, 2019) and to determine their ballistic transport when connected to semi-infinite leads (Calzolari *et al.*, 2004; Lee *et al.*, 2005). In this latter case, the formalism of Green's functions is used to embed a conductor into the surrounding environment. In all cases, the building blocks are Hamiltonian matrix elements between the localized MLWFs that are used to construct, LEGOTM-like, either the desired non-self-consistent Hamiltonian of a much larger nanostructure, or the self-energies embedding the conductor of interest into semi-infinite leads. Importantly, any solid (or surface) can be viewed as an infinite (or semi-infinite, in the case of surfaces) stack of “principal layers” interacting only with neighboring layers (Lee and Joannopoulos, 1981a,b). In this way, the infinite-dimensional real-space Hamiltonian can be divided into finite-sized Hamiltonian matrices; for a bulk system (i.e., infinite and periodic) the only independent components are H_{00} and H_{01} , where the former represents the interaction between MLWFs located in the same principal layer, and the latter the interaction between orbitals in one principal layer and the next.

As discussed by Nardelli (Nardelli, 1999), one can consider a system composed of a conductor C connected to two semi-infinite leads, R and L (C , R , and L are in themselves composed by a finite or infinite number of principal layers). The conductance through C is related to the scattering properties of C itself via the Landauer formula (Landauer, 1970):

$$\mathcal{G}(E) = \frac{2e^2}{h} \mathcal{T}(E), \quad (39)$$

where $\mathcal{T}$ is the transmission function, $\mathcal{G}$ is the conductance, and $\mathcal{T}$ is the probability that an electron injected at one end of the conductor will transmit to the other end.

This transmission function can be expressed in terms of the Green's functions of the conductors and of its coupling to the leads (Lee and Joannopoulos, 1981a,b) as

$$\mathcal{T} = \text{Tr}(\Gamma_L G_C^r \Gamma_R G_C^a), \quad (40)$$

where $G_C^{\{r,a\}}$ are the retarded and advanced Green's functions of the conductor, respectively, and $\Gamma_{\{L,R\}}$ are functions that describe the coupling of the conductor to the two leads. To compute the Green's function of the conductor, one starts from the equation for the Green's function of the whole system:

$$(\epsilon - H)G = \mathbb{1} \quad (41)$$

where $\epsilon = E + i\eta$ with η arbitrarily small, and $\mathbb{1}$ is the identity matrix. The matrix representation of G in a WFs basis can be partitioned as

$$\begin{pmatrix} G_L & G_{LC} & G_{LCR} \\ G_{CL} & G_C & G_{CR} \\ G_{LRC} & G_{RC} & G_R \end{pmatrix} = \quad (42)$$

$$\begin{pmatrix} (\epsilon - H_L) & H_{LC} & 0 \\ H_{LC}^\dagger & (\epsilon - H_C) & H_{CR} \\ 0 & H_{CR}^\dagger & (\epsilon - H_R) \end{pmatrix}^{-1}.$$

We can then write the conductor Green's function as

$$G_C(E) = (\epsilon - H_C - \Sigma_L - \Sigma_R)^{-1}, \quad (43)$$

where the effect of the semi-infinite leads on the conductor is described by the self-energies $\Sigma_{L,R}$, and the coupling functions $\Gamma_{\{L,R\}}$ are obtained from the self-energies as

$$\Gamma_{\{L,R\}} = i[\Sigma_{\{L,R\}}^r - \Sigma_{\{L,R\}}^a]; \quad (44)$$

here, the advanced self-energy is the Hermitian conjugate of the retarded one, and we solve for the latter. Given that the semi-infinite leads are also made of principal layers, one can construct the self-energies as (Nardelli, 1999)

$$\begin{aligned} \Sigma_L &= H_{LC}^\dagger (\epsilon - H_{00}^L - (H_{01}^L)^\dagger \bar{T}_L)^{-1} H_{LC}, \\ \Sigma_R &= H_{CR} (\epsilon - H_{00}^R - H_{01}^R T_R)^{-1} H_{CR}^\dagger, \end{aligned} \quad (45)$$

where, e.g., H_{00} describes the intralayer interactions and H_{01} the interlayer couplings. The transfer matrices $\bar{T}_{L,R}$ and $T_{L,R}$, defined such that $G_{10} = \bar{T}G_{00}$ and $G_{00} = \bar{T}G_{10}$, are calculated following the iterative procedure by Lopez-Sancho et al. (Sancho et al., 1984). The only inputs are the matrix elements of the Hamiltonian H_{mn} in a localized representation; the accuracy of the results depends on having principal layers that do not couple beyond next neighbors, i.e., on having a well-localized basis. The knowledge of the conductor Green's

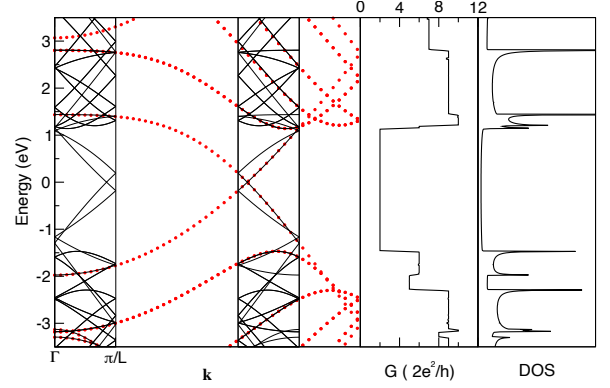


FIG. 5 Comparison of band structures calculated for a (5,5) single-walled carbon nanotube, either from MLWFs obtained from a supercell calculation (100 atoms, Γ sampling), and then Wannier-interpolated (solid thin line) or from a full diagonalization of the Hamiltonian (20-atom primitive cell, sampled with 5 k -points, red dots). The center and right panels show also the ballistic conductance and density of states calculated from the Green's functions (see text). Notably, only states at Γ are used in the disentanglement and Wannierization, where the system exhibits a pseudogap greater than 2 eV; still, this leads to a perfect description of the metallic bands and the van Hove singularities anywhere in the BZ, highlighting the role of MLWFs as excellent interpolators and building blocks for the non-self-consistent electronic structure of large-scale nanostructures (see also Lee et al. (2005)).

function G_C also gives direct information on the electronic spectrum via the spectral density of the electronic states: $N_C(E) = -(1/\pi)\text{Im}(\text{Tr}G_C(E))$.

As an example, we take a (5,5) single-wall carbon nanotube, described with a supercell containing 100 carbon atoms (five times the primitive cell); the disentanglement and Wannierization procedure results in 150 sp^2 orbitals and 100 p_z orbitals. Such supercell size is large enough to allow for Γ -sampling only in the BZ, and to have negligible overlap with the MLWFs belonging to the second next supercell. We show in Fig. 5 the results for the band structure, quantum conductance, and density of states, adapted from Lee et al. (2005). The metallic nanotube, notwithstanding its ~ 2 eV pseudo-gap at Γ , shows a band structure in perfect agreement with that obtained from a very fine sampling of the BZ, and perfect agreement between the steps in conductance and the peaks in the van Hove singularities, with the maxima and the minima of the band dispersions at arbitrary points in the BZ. While these methodologies based on WFs have been implemented either as an extension of Wannier90 (Shelley et al., 2011) or standalone in the WF-based WanT code (Ferretti et al., 2012), analogous techniques have been developed for other types of localized orbitals and implemented in simulation codes such as TranSiesta/Atomistix Toolkit (Brandbyge et al.,

2002), SMEAGOL (Rocha *et al.*, 2006), OpenMX (Ozaki *et al.*, 2010), NEMO5 (Fonseca *et al.*, 2013), nextnano (Birner *et al.*, 2007), NanoTCAD Vides (Bruzzone *et al.*, 2014; Fiori and Iannaccone, 2007), KWANT (Groth *et al.*, 2014), some of which also provide an interface with the Wannier ecosystem.

E. Berryology

1. Motivation

Berry phases and related quantities are central to the description of the electronic properties of crystals (Vanderbilt, 2018; Xiao *et al.*, 2010). Here are some representative examples.

1. The electronic contribution to the electric polarization of an insulator is given by

$$\mathbf{P}_{\text{el}} = -e \sum_n^{\text{occ}} \int_{\text{BZ}} \frac{d^3k}{(2\pi)^3} \mathbf{A}_{\mathbf{k},nn}, \quad (46)$$

where $-e$ is the electron charge and $\mathbf{A}_{\mathbf{k},nn}$ are diagonal elements of the Berry connection matrix,

$$\mathbf{A}_{\mathbf{k},mn} = i \langle u_{\mathbf{m}\mathbf{k}} | \nabla_{\mathbf{k}} u_{n\mathbf{k}} \rangle. \quad (47)$$

2. Off-diagonal elements of $\mathbf{A}_{\mathbf{k}}$ describe electric-dipole transition moments, allowing the interband optical conductivity to be expressed as

$$\sigma_{ab}(\omega) = \frac{ie^2}{\hbar} \sum_{m,n} \int_{\text{BZ}} \frac{d^3k}{(2\pi)^3} (f_{m\mathbf{k}} - f_{n\mathbf{k}}) \times \\ \times \frac{\varepsilon_{m\mathbf{k}} - \varepsilon_{n\mathbf{k}}}{\varepsilon_{m\mathbf{k}} - \varepsilon_{n\mathbf{k}} - \hbar(\omega + i0^+)} A_{\mathbf{k},nm}^a A_{\mathbf{k},mn}^b, \quad (48)$$

where $f_{n\mathbf{k}}$ is the Fermi-Dirac occupation factor.

3. The Berry curvature is defined as the curl of the Berry connection,

$$\Omega_{n\mathbf{k}} = \nabla_{\mathbf{k}} \times \mathbf{A}_{\mathbf{k},nn} = -\text{Im} \langle \nabla_{\mathbf{k}} u_{n\mathbf{k}} | \times | \nabla_{\mathbf{k}} u_{n\mathbf{k}} \rangle, \quad (49)$$

and its integral over the occupied states gives the intrinsic AHC,

$$\sigma_{yx} = \frac{e^2}{\hbar} \int_{\text{BZ}} \frac{d^3k}{(2\pi)^3} \sum_n f_{n\mathbf{k}} \Omega_{n\mathbf{k}}^z. \quad (50)$$

4. The ground-state orbital magnetization reads

$$\mathbf{M}_{\text{orb}} = \int_{\text{BZ}} \frac{d\mathbf{k}}{(2\pi)^3} \sum_n f_{n\mathbf{k}} \left[\mathbf{m}_{n\mathbf{k}}^{\text{orb}} + \frac{e}{\hbar} (\varepsilon_{\text{F}} - \varepsilon_{n\mathbf{k}}) \Omega_{n\mathbf{k}} \right], \quad (51)$$

with ε_{F} the Fermi energy and

$$\mathbf{m}_{n\mathbf{k}}^{\text{orb}} = \frac{e}{2\hbar} \text{Im} \langle \nabla_{\mathbf{k}} u_{n\mathbf{k}} | \times \left(\hat{H}_{\mathbf{k}} - \varepsilon_{n\mathbf{k}} \right) | \nabla_{\mathbf{k}} u_{n\mathbf{k}} \rangle \quad (52)$$

the intrinsic orbital moment of a Bloch state.

5. To first order in applied fields $\mathbf{E}$ and $\mathbf{B}$, the semi-classical equations of motion for a wavepacket in a Bloch band read

$$\dot{\mathbf{r}} = \frac{1}{\hbar} \nabla_{\mathbf{k}} \tilde{\varepsilon}_{n\mathbf{k}} - \dot{\mathbf{k}} \times \Omega_{n\mathbf{k}}, \quad (53a)$$

$$\dot{\mathbf{k}} = -\frac{e}{\hbar} \mathbf{E} - \frac{e}{\hbar} \dot{\mathbf{r}} \times \mathbf{B}, \quad (53b)$$

where

$$\tilde{\varepsilon}_{n\mathbf{k}} = \varepsilon_{n\mathbf{k}} - \left(\mathbf{m}_{n\mathbf{k}}^{\text{spin}} + \mathbf{m}_{n\mathbf{k}}^{\text{orb}} \right) \cdot \mathbf{B} \quad (54)$$

is the Zeeman-shifted band energy.

The motivation to apply Wannier interpolation to Berry-type quantities came from pioneering *ab initio* calculations of the AHC in the ferromagnets SrRuO₃ (Fang *et al.*, 2003) and BCC Fe (Yao *et al.*, 2004), which revealed the integrand of Eq. (50) to be strongly peaked in the vicinity of avoided crossings between occupied and empty bands; resulting in the need to sample the BZ over millions of k -points to reach convergence. An efficient Wannier-interpolation scheme for evaluating the AHC was developed in Wang *et al.* (2006), and since then the methodology has been applied to many other properties.

Wannier interpolation of Berry-type quantities was introduced in version 2 of Wannier90 as part of its post-processing code `postw90` (Mostofi *et al.*, 2014), with the ability to compute AHC (Wang *et al.*, 2006), interband optical conductivity (Yates *et al.*, 2007), and orbital magnetization (Lopez *et al.*, 2012). The list of available properties has grown considerably since then, and more recently the WannierBerri code package (Tsirkin, 2021) introduced several methodological improvements including “pruned FFT” (Markel, 1971; Sorensen and Burrus, 1993) (a combination of fast and slow Fourier transforms), and the use of symmetries and of the tetrahedron method for BZ integrals. Other codes, including WannierTools (Wu *et al.*, 2018) and `linres` (Železný, 2023), implement some functionalities but with additional approximations (see discussion of Eq. (60) below). The reader should consult the documentation of the codes for an up-to-date description of their capabilities.

In the following, we outline the basic interpolation strategy for Berry-type quantities, using the off-diagonal Berry connection as an example (the diagonal Berry connection that enters the Berry phase requires a separate treatment, see Sec. III.F). For discussion purposes, we adopt the modified phase convention for Bloch sums indicated in Eq. (31).

2. Wannier interpolation of the interband Berry connection

We wish to evaluate off-diagonal elements of the Berry connection matrix in the Hamiltonian gauge. Inserting

in Eq. (47) the relation (18) between interpolated Bloch states in the Hamiltonian and Wannier gauges, we obtain

$$\mathbf{A}_{\mathbf{k}'}^{\text{H}} = i\mathcal{U}_{\mathbf{k}'}^\dagger \nabla_{\mathbf{k}} \mathcal{U}_{\mathbf{k}'} + \mathcal{U}_{\mathbf{k}'}^\dagger \mathbf{A}_{\mathbf{k}'}^{\text{W}} \mathcal{U}_{\mathbf{k}'} . \quad (55)$$

Note the extra (first) term compared to the gauge-transformation rule (25) for a k -local matrix object. Recalling from Eq. (14) that the columns of $\mathcal{U}_{\mathbf{k}'}$ are eigenvectors of $H_{\mathbf{k}'}^{\text{W}}$, the off-diagonal matrix elements of that term can be evaluated from non-degenerate perturbation theory as

$$\left(\mathcal{U}_{\mathbf{k}'}^\dagger \nabla_{\mathbf{k}} \mathcal{U}_{\mathbf{k}'} \right)_{mn} = \frac{\left[\mathcal{U}_{\mathbf{k}'}^\dagger (\nabla_{\mathbf{k}} H_{\mathbf{k}'}^{\text{W}}) \mathcal{U}_{\mathbf{k}'} \right]_{mn}}{\varepsilon_{n\mathbf{k}'}^{\text{H}} - \varepsilon_{m\mathbf{k}'}^{\text{H}}} . \quad (56)$$

All quantities on the right-hand side can be obtained from Eqs. (13) and (14) starting from $\langle 0i|\hat{H}|\mathbf{R}j\rangle$ and $\boldsymbol{\tau}_j = \langle 0j|\hat{\mathbf{r}}|0j\rangle$ (the latter appears in the modified phase factors in Eq. (31)). For the second term in Eq. (55), we also need

$$\mathbf{A}_{\mathbf{k}',ij}^{\text{W}} = \sum_{\mathbf{R}} e^{i\mathbf{k}' \cdot (\mathbf{R} + \boldsymbol{\tau}_j - \boldsymbol{\tau}_i)} \mathbf{d}_{ij}(\mathbf{R}) , \quad (57)$$

which follows from inserting in Eq. (47) the Bloch sum (16) with the modified phase factors. Here $\mathbf{d}_{ij}(\mathbf{R})$ are the off-diagonal matrix elements of $\hat{\mathbf{r}}$ in the Wannier basis, that is,

$$\langle 0i|\hat{\mathbf{r}}|\mathbf{R}j\rangle = \delta_{\mathbf{R},0} \delta_{ij} \boldsymbol{\tau}_j + \mathbf{d}_{ij}(\mathbf{R}) . \quad (58)$$

The matrix elements $\langle 0i|\hat{H}|\mathbf{R}j\rangle$ are evaluated using Eq. (12), and the corresponding procedure for $\langle 0i|\hat{\mathbf{r}}|\mathbf{R}j\rangle$ is as follows. First, we can use Eq. (5a) to write $\langle 0i|\hat{\mathbf{r}}|\mathbf{R}j\rangle$ as $(1/N) \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{R}} A_{\mathbf{k},ij}^{\text{W}}$. Since the Bloch functions are smooth in the Wannier gauge, $\mathbf{A}_{\mathbf{k},ij}^{\text{W}}$ can be evaluated on the *ab initio* grid by discretizing the $\mathbf{k}$ derivative appearing in Eq. (47). Adopting the finite-differences scheme described in Marzari and Vanderbilt (1997) and Mostofi *et al.* (2008) we obtain

$$\begin{aligned} \langle 0i|\hat{\mathbf{r}}|\mathbf{R}j\rangle &= \frac{i}{N} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{R}} \sum_{\mathbf{b}} w_b \mathbf{b} \times \\ &\times \sum_{m,n} V_{\mathbf{k},mi}^* M_{mn}^{(\mathbf{k},\mathbf{b})} V_{\mathbf{k}+\mathbf{b},nj} , \end{aligned} \quad (59)$$

where $\mathbf{b}$ are vectors connecting neighboring grid points, w_b are appropriately chosen weights, $M^{(\mathbf{k},\mathbf{b})}$ are the overlap matrices defined by Eq. (8), and $V_{\mathbf{k}}$ are the Wannierization matrices in Eq. (5b); since the overlap matrices were computed in preparation for the WF construction procedure, and the Wannierization matrices were obtained at the end of that procedure, both are readily available.

Once $\langle 0i|\hat{H}|\mathbf{R}j\rangle$ and $\langle 0i|\hat{\mathbf{r}}|\mathbf{R}j\rangle$ have been tabulated, the interband Berry connection can be evaluated from Eqs. (55)–(57), with the matrices $\mathcal{U}_{\mathbf{k}'}$ therein (along with

the interpolated energy eigenvalues) given by Eq. (14). Finally, the Berry connection and energy eigenvalues are inserted in Eq. (48) to obtain the interband optical conductivity (Yates *et al.*, 2007).

Equation (59) entails a numerical error of order $(\Delta k)^2$, where Δk is the *ab initio* mesh spacing (Marzari and Vanderbilt, 1997; Mostofi *et al.*, 2008). The direct real-space mesh integration method mentioned below Eq. (12) should be free of such errors, but it is not as practical in the context of k -space Wannierization schemes. It is therefore desirable to develop improved discretized k -space formulas for $\langle 0i|\hat{\mathbf{r}}|\mathbf{R}j\rangle$ and related matrix elements. A higher-order generalization of the discretization scheme of Marzari and Vanderbilt (1997) and Mostofi *et al.* (2008) was recently introduced (Cistaro *et al.*, 2023), and further improvements are currently under way (Ghim, 2022; Lihm, 2022).

In empirical tight-binding, it is customary to approximate the position matrix elements by dropping the second term in Eq. (58) (Foreman, 2002; Vanderbilt, 2018),

$$\langle 0i|\hat{\mathbf{r}}|\mathbf{R}j\rangle \approx \delta_{\mathbf{R},0} \delta_{ij} \boldsymbol{\tau}_j . \quad (60)$$

In this approximation, and when the modified phase convention (31) is used, the matrix $\mathbf{A}_{\mathbf{k}}^{\text{W}}$ in Eq. (57) vanishes; Eq. (55) for $\mathbf{A}_{\mathbf{k}}^{\text{H}}$ then reduces to its first term, which can be interpreted as the “internal” Berry connection of the tight-binding eigenvectors; this is how tight-binding codes such as PythTB evaluate Berry phases and curvatures (PythTB, 2023b), which is quite natural in the context of toy-model calculations.

The above approximation is harder to justify when using *ab initio* WFs, given that the discarded $\mathbf{d}_{ij}(\mathbf{R})$ matrix elements are readily available, as mentioned above; even so, that approximation is made by some codes, including WannierTools (Wu *et al.*, 2018) and linres (Železný, 2023). The role of intra-atomic $\mathbf{d}_{ij}(\mathbf{R})$ matrix elements in tight-binding calculations of the linear dielectric function was studied in Pedersen *et al.* (2001); in Ibañez-Azpiroz *et al.* (2022), that analysis was extended to inter-atomic matrix elements and to non-linear optical responses, using Wannier interpolation.

3. Other Berry-type quantities

The interpolation of the Berry curvature $\boldsymbol{\Omega}_{\mathbf{k}}$ proceeds along similar lines, allowing to compute the AHC from Eq. (50) (Wang *et al.*, 2006), and the procedure can be extended to spin Hall conductivity (SHC) (Qiao *et al.*, 2018; Ryoo *et al.*, 2019) or non-linear responses. For example, non-linear optical and AHCs involve k -derivatives of $\mathbf{A}_{\mathbf{k}}$ and $\boldsymbol{\Omega}_{\mathbf{k}}$, respectively (Aversa and Sipe, 1995; Sodemann and Fu, 2015); in the same way as band derivatives (see Sec. III.C.2), both are conveniently evaluated by Wannier interpolation (Ibañez Azpiroz *et al.*, 2018; Liu *et al.*, 2023; Wang *et al.*, 2017). To interpolate $\mathbf{A}_{\mathbf{k}}$, $\boldsymbol{\Omega}_{\mathbf{k}}$,

and their k -derivatives, only $\langle 0i|\hat{H}|\mathbf{R}j\rangle$ and $\langle 0i|\hat{\mathbf{r}}|\mathbf{R}j\rangle$ are needed, but other quantities require additional matrix elements. For example, the orbital moment in Eq. (52) requires $\langle 0i|\hat{H}(\hat{\mathbf{r}}-\mathbf{R})|\mathbf{R}j\rangle$ and $\langle 0i|\hat{\mathbf{r}}_a\hat{H}(\hat{\mathbf{r}}-R)_b|\mathbf{R}j\rangle$ (Lopez *et al.*, 2012); and while the former can be evaluated on the *ab initio* grid using the same ingredients entering Eqs. (12) and (59) for $\langle 0i|\hat{H}|\mathbf{R}j\rangle$ and $\langle 0i|\hat{\mathbf{r}}|\mathbf{R}j\rangle$, the latter also requires $\langle u_{m\mathbf{k}+\mathbf{b}_1}|\hat{H}_{\mathbf{k}}|u_{n\mathbf{k}+\mathbf{b}_2}\rangle$, which must be calculated separately (Lopez *et al.*, 2012). As in the case of Eq. (55) for $\mathbf{A}_{\mathbf{k}'}$, the resulting interpolation formula contains an “internal” term analogous to Eq. (52) itself, but expressed in terms of the TB eigenvectors and Hamiltonian. That term can be evaluated from $\langle 0i|\hat{H}|\mathbf{R}j\rangle$ and $\boldsymbol{\tau}_j$ alone, but it leaves out important atomic-like contributions that are encoded in the other matrix elements needed for a full calculation (Lopez *et al.*, 2012; Nikolaev and Solovyev, 2014).

The *ab initio* matrix elements needed for various interpolation tasks are listed in Table I. The files `*.eig` and `*.mmn` are required already for constructing the WFs, and therefore they are provided by the interface code of every *ab initio* code that is compatible with Wannier90. The file `*.spn` is provided by the interface of both Quantum ESPRESSO and VASP. As for the other matrix elements listed in Table I, at present they are only implemented in `pw2wannier90.x`, the interface of Quantum ESPRESSO.

As a workaround for obtaining these quantities from the output of other *ab initio* engines, one can resort to a sum-over-states procedure. For example, the uHu matrix elements may be expressed as

$$\langle u_{m\mathbf{k}+\mathbf{b}_1}|\hat{H}_{\mathbf{k}}|u_{n\mathbf{k}+\mathbf{b}_2}\rangle \approx \sum_l^{l_{\max}} \langle u_{m\mathbf{k}+\mathbf{b}_1}|u_{l\mathbf{k}}\rangle \varepsilon_{l\mathbf{k}} \langle u_{l\mathbf{k}}|u_{n\mathbf{k}+\mathbf{b}_2}\rangle \quad (61)$$

in terms of the energy eigenvalues and overlap matrices, and the relevant matrices for spin Hall conductivity can be obtained similarly (Qiao *et al.*, 2018). Since the summation is done before Wannierization, the number $l_{\max}$ of states included in the non-selfconsistent *ab initio* calculation can be systematically increased until the desired level of convergence is reached.

The above procedure is implemented for uHu, uIu, sHu and sIu in the utility `mmn2uHu` (WannierBerri, 2023) provided with the WannierBerri code package (Tsirkin, 2021). Besides its use as a workaround, it can serve as a benchmark for testing future implementations of those matrix elements in various interface codes between *ab initio* and Wannier engines.

F. Topological invariants and related properties

The topological aspects of band theory have been studied intensively over the past two decades (Hasan and Kane, 2010; Vanderbilt, 2018), and *ab initio* calculations have been central to that effort (Wang and Zhang, 2017):

they are used for identifying candidate topological materials, to determine topological invariants, and to calculate surface bands that can be compared with angle-resolved photoemission measurements.

WFs feature prominently in topological band theory. For example, quantum anomalous Hall insulators (Hal-dane, 1988) (a.k.a. Chern insulators) can be defined as 2D systems where it is not possible¹ to construct a set of exponentially-localized WFs (Brouder *et al.*, 2007; Thonhauser and Vanderbilt, 2006; Vanderbilt, 2018): this is known as a “topological obstruction”. More generally, symmetry-protected topological insulators can typically be defined as insulators for which it is not possible to construct a set of WFs spanning the valence bands without breaking the protecting symmetry in the choice of gauge (Bradlyn *et al.*, 2017; Soluyanov and Vanderbilt, 2011a). Prominent examples include 2D quantum spin Hall insulators (Bernevig and Zhang, 2006; Kane and Mele, 2005a,b) and 3D $\mathbb{Z}_2$ topological insulators (Fu *et al.*, 2007), where the protecting symmetry is time reversal, and topological crystalline insulators, which are instead protected by crystalline symmetries (Fu, 2011).

A second example is the “Wannier spectrum” defined by the centers of hybrid (a.k.a. hermaphrodite) orbitals (Sgiarovello *et al.*, 2001) that are Wannier-like along $\hat{\mathbf{z}}$ and Bloch-like along $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$. The surface energy spectrum $\varepsilon_n(k_x, k_y)$ can be continuously deformed into the bulk Wannier spectrum $z_n(k_x, k_y)$ obtained by Wannierizing along the surface normal (Fidkowski *et al.*, 2011; Neupert and Schindler, 2018), allowing to infer the topological flow of the surface energy bands from that of the bulk Wannier bands (Taherinejad *et al.*, 2014). In some cases, the topological indices can be deduced from the Wannier band structure (Gresch *et al.*, 2017; Varnava *et al.*, 2020).

WFs also play a more practical role in the study of topological materials, as several of the electronic-structure packages that are commonly used to characterize them rely on a Wannier/TB representation (despite the obstruction mentioned above (Bradlyn *et al.*, 2017; Soluyanov and Vanderbilt, 2011a), topological insulators still afford a Wannier representation, provided that the WFs span a few low-lying conduction states along with the valence bands).

In particular, both PythTB (PythTB, 2023a) and WannierTools (Wu *et al.*, 2018) work with orthogonal TB models, and have the option to import TB Hamiltonians generated by a Wannier engine using either the `*.hr.dat` or the `*.tb.dat` file format (as mentioned above, the latter also includes the matrix elements of

¹ Very recent work (Gunawardana *et al.*, 2023) suggests that it might be possible to construct optimally localized WFs even in presence of non-vanishing Chern numbers, at least for the simple case of one isolated band.

Matrix element	Wannier90 file	Needed for	Implemented in	Computed by <code>mmn2uHu</code> from
$\langle u_{m\mathbf{k}} \hat{H}_{\mathbf{k}} u_{n\mathbf{k}} \rangle = \varepsilon_{n\mathbf{k}} \delta_{mn}$	<code>*.eig</code>	Transport, optics, $\mathbf{m}_{\mathbf{k}}^{\text{orb}}$	All	
$\langle u_{m\mathbf{k}} \hat{\sigma} u_{n\mathbf{k}} \rangle$	<code>*.spn</code>	$\mathbf{m}_{\mathbf{k}}^{\text{spin}}$	QE, VASP	
$\langle u_{m\mathbf{k}} u_{n\mathbf{k}+\mathbf{b}} \rangle$	<code>*.mmn</code>	$\mathbf{A}_{\mathbf{k}}, \mathbf{\Omega}_{\mathbf{k}}, \mathbf{m}_{\mathbf{k}}^{\text{orb}}$, all below	All	
$\langle u_{m\mathbf{k}+\mathbf{b}_1} \hat{H}_{\mathbf{k}} u_{n\mathbf{k}+\mathbf{b}_2} \rangle$	<code>*.uHu</code>	$\mathbf{m}_{\mathbf{k}}^{\text{orb}}$	QE	<code>*.eig, *.mmn</code>
$\langle u_{m\mathbf{k}+\mathbf{b}_1} u_{n\mathbf{k}+\mathbf{b}_2} \rangle$	<code>*.uIu</code>	$q_{\mathbf{k}}^{ab}$	QE	<code>*.mmn</code>
$\langle u_{m\mathbf{k}} \hat{\sigma} \hat{H}_{\mathbf{k}} u_{n\mathbf{k}+\mathbf{b}} \rangle$	<code>*.sHu</code>	} SHC in Ryoo et al. (2019)	QE	<code>*.eig, *.mmn, *.spn</code>
$\langle u_{m\mathbf{k}} \sigma u_{n\mathbf{k}+\mathbf{b}} \rangle$	<code>*.sIu</code>		QE	<code>*.mmn, *.spn</code>

TABLE I *Ab initio* matrix elements that are used *explicitly* in setting up the WF matrix elements needed to perform common interpolation tasks, SHC stands for spin Hall conductivity. * denotes the seedname specified in the input file of `Wannier90`. QE = `Quantum ESPRESSO`, $\mathbf{A}_{\mathbf{k}}$ is the Berry connection or electric dipole matrix (47), $\mathbf{\Omega}_{\mathbf{k}}$ is the Berry curvature (49), $\mathbf{m}_{\mathbf{k}}^{\text{orb}}$ is the intrinsic orbital magnetic dipole (52), and $q_{\mathbf{k}}^{ab}$ is the intrinsic electric quadrupole (for the matrix definitions of $\mathbf{m}_{\mathbf{k}}^{\text{orb}}$ and $q_{\mathbf{k}}^{ab}$, see [Óscar Pozo Ocaña and Souza \(2023\)](#)). The optical conductivity (48) in the electric-dipole approximation involves $\varepsilon_{n\mathbf{k}}$ and $\mathbf{A}_{\mathbf{k}}$, the AHC (50) involves $\varepsilon_{n\mathbf{k}}$ and $\mathbf{\Omega}_{\mathbf{k}}$, and the orbital magnetization (51) involves $\varepsilon_{n\mathbf{k}}$, $\mathbf{\Omega}_{\mathbf{k}}$ and $\mathbf{m}_{\mathbf{k}}^{\text{orb}}$. Spatially-dispersive responses such as natural optical activity depend on $\varepsilon_{n\mathbf{k}}$, $\mathbf{A}_{\mathbf{k}}$, $\mathbf{m}_{\mathbf{k}}^{\text{orb}}$, $\mathbf{m}_{\mathbf{k}}^{\text{spin}}$, and $q_{\mathbf{k}}^{ab}$ ([Óscar Pozo Ocaña and Souza, 2023](#)). The `mmn` (and `eig`) matrix elements are needed for constructing the (disentangled) WFs, and hence they are used *implicitly* when interpolating any physical quantity.

the position operator in the Wannier basis, and the coordinates of the lattice vectors). The key point is that the Wannierized Hamiltonian preserves the topological features of the original first-principles electronic structure; the identification and characterization of those features can therefore be carried out entirely in the Wannier representation, which is often more convenient and/or efficient than proceeding directly from the *ab initio* Bloch states.

The simplest example of a topological band-structure feature is an isolated touching between a pair of bands, known as a “Weyl point” ([Armitage et al., 2018](#); [Vanderbilt, 2018](#)). Weyl points are fundamentally different from weak avoided crossings, but most band interpolation schemes are unable to tell them apart; instead, Wannier interpolation correctly distinguishes between the two. The distinction is rooted in the fact that a Weyl node acts as a monopole source or sink of Berry curvature in $\mathbf{k}$ space, allowing to associate with it a topological invariant known as the “chiral charge.”

The chiral charge χ (typically ± 1 , but sometimes ± 2 or ± 3 ([Fang et al., 2012](#); [Tsirkin et al., 2017](#))) can be determined in two different ways: (i) from the quantized Berry-curvature flux through a small surface $\mathcal{S}$ enclosing the Weyl point ([Gosálbez-Martínez et al., 2015](#); [Vanderbilt, 2018](#)),

$$\int_{\mathcal{S}} \mathbf{\Omega}_{n\mathbf{k}} \cdot \hat{\mathbf{n}} = -2\pi\chi, \quad (62)$$

where $\hat{\mathbf{n}}$ is a unit vector in the direction of $\nabla_{\mathbf{k}}\varepsilon_{n\mathbf{k}}$; (ii) by evaluating the Berry phase

$$\phi_n(\mathcal{C}) = \oint_{\mathcal{C}} \mathbf{A}_{n\mathbf{k}} \cdot d\mathbf{k} \quad (63)$$

around contours $\mathcal{C}$ at fixed latitude on an enclosing spherical surface, and then tracking its evolution from zero at the south pole to $2\pi\chi$ at the north pole ([Gresch et al., 2017](#)).

The latter procedure is implemented in both `Z2Pack` ([Z2pack, 2023](#)) and `WannierTools`. All that is required is the TB Hamiltonian $\langle \mathbf{0}i | \hat{H} | \mathbf{R}j \rangle$, from which one obtains the eigenvectors on a discrete mesh $\{\mathbf{k}_j\}$ of points along each contour; the Berry phase is then evaluated by finite differences from the overlaps between TB eigenvectors on consecutive points along $\mathcal{C}$ as follows ([Vanderbilt, 2018](#)),

$$\phi_n^{(\text{int})}(\mathcal{C}) = -\text{Im} \ln \Pi_j \langle \phi_{n\mathbf{k}_j} | \phi_{n\mathbf{k}_{j+1}} \rangle, \quad (64)$$

where $\|\phi_{n\mathbf{k}_j}\rangle$ denotes a column vector of the matrix $\mathcal{U}_{\mathbf{k}_j}$ defined by Eq. (14). The above expression corresponds, in the language of Sec. III.E, to the internal part of the Berry phase (63), which also contains an external part

$$\phi_n^{(\text{ext})}(\mathcal{C}) = \sum_j \langle \phi_{n\mathbf{k}_j} | \mathbf{A}_{\mathbf{k}}^W | \phi_{n\mathbf{k}_j} \rangle \cdot \Delta\mathbf{k}, \quad (65)$$

where $\Delta\mathbf{k} = (\mathbf{k}_{j+1} - \mathbf{k}_{j-1})/2$ ([Wang et al., 2007](#)). The two parts arise from discretizing the integral along $\mathcal{C}$ of the two terms in Eq. (55) for the interpolated Berry connection; the internal term only depends on $\langle \mathbf{0}i | \hat{H} | \mathbf{R}j \rangle$, while the external one also requires $\langle \mathbf{0}i | \hat{\mathbf{r}} | \mathbf{R}j \rangle$. The 2π indeterminacy in the Berry phase comes from the former, while the latter is single-valued and hence it does not contribute to the quantized change in Berry phase from the south to the north pole of a spherical surface; this is why χ can be determined from the TB Hamiltonian alone.

Weyl crossings can occur at arbitrary points in the BZ, which makes it difficult to spot them in the band structure. By allowing to quickly evaluate energy eigenvalues and band velocities at arbitrary k -points, Wannier interpolation provides a practical solution to this problem ([Gosálbez-Martínez et al., 2015](#)): to locate the degeneracies between bands n and $n+1$, define a gap function $\varepsilon_{n+1\mathbf{k}} - \varepsilon_{n\mathbf{k}}$, and search for its minima using a

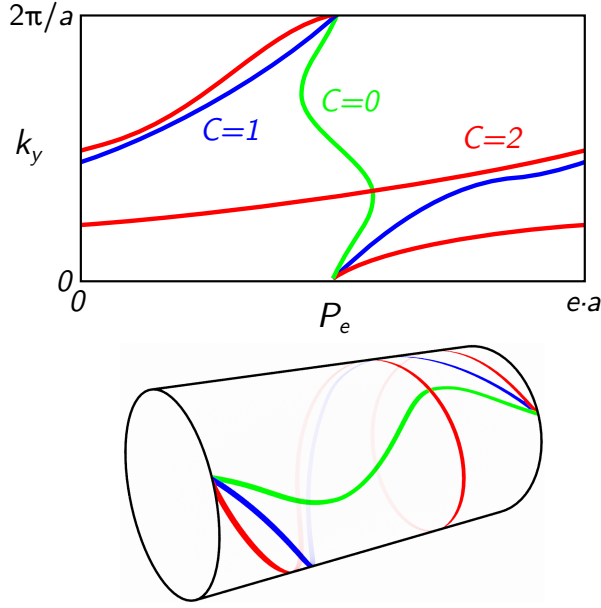


FIG. 6 Sketch of some possible evolutions of hybrid polarization $P_e(k_y)$, i.e. the sum of hybrid Wannier charge centers, across the BZ. Chern numbers C correspond to different winding numbers. See [Gresch et al. \(2017\)](#) for an in-depth discussion.

minimization method such as conjugate-gradient, starting from a sufficiently dense grid of k -points. After discarding local minima where the gap function is above some numerical threshold, one is left with candidate degeneracies that can be further characterized; they include not only point nodes such as Weyl and Dirac nodes ([Armitage et al., 2018](#)), but also nodal lines ([Fang et al., 2016](#); [Yang et al., 2018](#)). This procedure is implemented in **WannierTools**.

Topological materials feature characteristic boundary modes that reflect the bulk topology ([Hasan and Kane, 2010](#); [Vanderbilt, 2018](#)). In the case of Weyl semimetals, those modes take the form of “Fermi arcs” connecting the projections of bulk Weyl nodes of opposite chirality onto the surface BZ ([Armitage et al., 2018](#); [Vanderbilt, 2018](#)). **PythTB** and **WannierTools** allow to terminate a bulk TB model along specified directions, creating ribbons or slabs whose boundary modes can then be inspected by plotting the energy bands. **WannierTools** also has the option to obtain a surface spectral function from the Green’s function calculated for a semi-infinite system.

As already mentioned, another very useful tool for diagnosing topological behaviors is a hybrid representation of the electronic structure in terms of orbitals that are localized in one spatial direction only, remaining extended in the others ([Sgierovello et al., 2001](#)). To define these hybrid Wannier functions (HWFs) for lattices of arbitrary symmetry, it is convenient to work in reduced coordinates. Consider a 2D crystal, and let $\mathbf{k} = k_1\mathbf{b}_1 + k_2\mathbf{b}_2$ and $\hat{\mathbf{r}} = \hat{x}_1\mathbf{a}_1 + \hat{x}_2\mathbf{a}_2$. Choosing $\mathbf{b}_2$ as the localization

direction, the HWFs are defined as

$$|h_{k_1ln}\rangle = \frac{1}{N_2} \sum_{k_2} e^{-i2\pi k_2 l} |\psi_{\mathbf{nk}}\rangle, \quad (66)$$

where N_2 is the number of distinct values of k_2 in the BZ, and l labels cells along $\mathbf{a}_2$. The topological indices can be determined from the hybrid Wannier centers

$$x_{2,k_1ln} = \langle h_{k_1ln} | \hat{x}_2 | h_{k_1ln} \rangle; \quad (67)$$

for bulk materials, the analysis is carried out on high-symmetry BZ planes. This hybrid-Wannier (or “Wilson-loop”) scheme is implemented in **Z2Pack**, and a detailed description of the methodology can be found in [Gresch et al. \(2017\)](#). In **Z2Pack**, the hybrid Wannier centers (67) are obtained from a parallel-transport construction, starting from the overlap matrices (8) ([Taherinejad et al., 2014](#)). The same procedure is implemented in **PythTB** and **WannierTools** for Wannier/TB Hamiltonians; the needed overlaps are then taken between TB eigenstates, as in Eq. (64) above (**Z2Pack** can also operate in this mode). We remark that **Z2Pack** ([Gresch et al., 2017](#)) allows to work both with Wannier/TB Hamiltonians, and directly with first-principles engines such as **Quantum ESPRESSO** and **VASP**.

In closing, it is worth bearing in mind the different design philosophies of the three packages surveyed in this section. As already mentioned in Sec. III.C, **PythTB** ([PythTB, 2023a](#)) was designed with TB “toy models” in mind, and to be used as a pedagogical tool; it allows to compute several geometric and topological quantities (Berry phases and curvatures, Chern numbers, hybrid Wannier centers), and to generate ribbons and slabs to expose their boundary modes. Although **PythTB** can also import large Wannier models (which can be truncated internally), the code is not optimized for speed; however, a high-performance Numba ([Lam et al., 2015](#)) implementation of **PythTB** better suited for that purpose is available ([Numba-PythTB, 2023](#)). **WannierTools** ([Wu et al., 2018](#)), on the other hand, is primarily designed to work with large Wannier models, and is parallelized using MPI. Its distinctive features include searching for band degeneracies, and plotting surface spectral functions. Finally, **Z2Pack** is focused on the HWF scheme; it is not primarily a “post-Wannier” code, since it can circumvent the need to use a Wannier engine by directly reading the *ab initio* overlap matrices, which can help streamline high-throughput calculations ([Grassano et al., 2023](#); [Marrazzo et al., 2019](#)).

While topological invariants are generally introduced for crystalline periodic systems in PBCs, there are a number of scenarios that require either the use of open boundary conditions (OBCs) or the adoption of large supercells with Γ -only sampling; in both cases, standard approaches are of no avail. Relevant examples include the study of Anderson ([Groth et al., 2009](#); [Jiang et al.,](#)

2009; Li *et al.*, 2009) and amorphous topological insulators (Corbae *et al.*, 2023), heterogeneous systems such as trivial/topological junctions (Bianco and Resta, 2011), molecular dynamics simulations, and any other use case that does not fit a small primitive cell with BZ sampling. Among many other approaches (see Corbae *et al.* (2023) for a dedicated overview), single-point sampling (Ceresoli and Resta, 2007; Favata and Marrazzo, 2023) and local markers (Bianco and Resta, 2011) have been introduced to study topology for non-crystalline systems.

Thanks to the use of WFs as a basis set, these techniques can be implemented seamlessly in the same framework both for model TB (like the Haldane (Haldane, 1988) or the Kane–Mele models (Kane and Mele, 2005a,b)) and *ab initio* TB, where the latter is obtained by constructing WFs on top of any electronic structure calculations including DFT, GW or DMFT (as discussed in Sec. III.C). This strategy has been adopted by the **StrawBerryPy** code (Baù and Marrazzo, 2023; Favata and Marrazzo, 2023; StrawBerryPy, 2023), where model or *ab initio* TB Hamiltonians are read and manipulated either through **PythTB** or **TBmodels**, allowing to calculate a few single-point and local topological invariants such as the Chern number and the $\mathbb{Z}_2$ invariant, as well as other quantum-geometrical quantities of the electronic structure that are relevant for topological materials.

G. Electron-phonon interactions

1. Methodology

In the past decade, we have witnessed a community-wide effort to develop advanced computational approaches and simulation tools for atomistic modeling of function-defining properties of materials. A primary focus of this ongoing research has been the accurate description of electron-phonon (e-ph) interactions from first-principles (Giustino, 2017), as they determine many materials properties of technological interest such as electrical and thermal transport (Brunin *et al.*, 2020a,b; Chaves *et al.*, 2020; Cheng *et al.*, 2020; Jhalani *et al.*, 2020; Li, 2015; Macheda *et al.*, 2022; Macheda and Bonini, 2018; Maliyov *et al.*, 2021; Park *et al.*, 2020; Poncé *et al.*, 2020; Poncé *et al.*, 2021, 2018, 2023; Protik and Broido, 2020; Protik and Kozinsky, 2020; Zhou *et al.*, 2021b), phonon-assisted light absorption (Bushick and Kioupakis, 2023; Noffsinger *et al.*, 2012), phonon-mediated superconductivity (Errea *et al.*, 2020; Heil *et al.*, 2017; Lafuente-Bartolome *et al.*, 2020; Lilia *et al.*, 2022; Margine and Giustino, 2013), and polaron formation (Falletta and Pasquarello, 2022; Lafuente-Bartolome *et al.*, 2022; Lee *et al.*, 2021; Sio and Giustino, 2023; Sio *et al.*, 2019; Zhou and Bernardi, 2019). A comprehensive review of the theory of e-ph interactions in solids from the point of view of *ab initio* calculations is given by

Giustino (Giustino, 2017). An important contribution to the e-ph problem has been recently made by Stefanucci and colleagues (Stefanucci *et al.*, 2023), who have developed an *ab initio* many-body quantum theory of electrons and phonons in and out-of-equilibrium.

There is a well established formalism for computing the e-ph matrix elements from first-principles using DFPT (Baroni *et al.*, 2001, 1987; Gonze, 1997; Savrasov, 1992). However, all aforementioned properties are notoriously difficult to evaluate with desired accuracy using DFPT calculations directly due to the prohibitive computational cost. To achieve numerical convergence, the e-ph matrix elements need to be computed on ultra-dense electron ($\mathbf{k}$) and phonon ($\mathbf{q}$) BZ grids with $10^6 - 10^7$ points. Specialized numerical techniques, such as linear (Brunin *et al.*, 2020b; Li, 2015) or Wannier (Calandra *et al.*, 2010; Giustino *et al.*, 2007a,b) interpolation and Fermi-surface harmonics (Allen, 1976; Lafuente-Bartolome *et al.*, 2020), have been developed to address this problem. In particular, the interpolation of the e-ph matrix elements using MLWFs (Marzari *et al.*, 2012) introduced by Giustino, Cohen, and Louie (Giustino *et al.*, 2007a) has proven very successful for enabling highly accurate and efficient calculations of e-ph interactions, and the approach has been implemented in a number of codes (Cepellotti *et al.*, 2022; Lee *et al.*, 2023; Marini *et al.*, 2023; Noffsinger *et al.*, 2010; Poncé *et al.*, 2016; Zhou *et al.*, 2021a).

We note that an alternative to the computation of the e-ph interaction with DFPT is offered by the finite displacement scheme in real space (Chaput *et al.*, 2019; Engel *et al.*, 2020, 2022). While this approach requires large supercells to reach convergence, it has the advantage of being universally applicable to any functional, including hybrid or metagradient functionals, as well as more complicated exchange-correlation potentials, where higher-order derivatives of the functional are not readily available.

Below, we focus on the DFPT approach and outline the interpolation procedure to compute the e-ph matrix elements on ultra-dense meshes using WFs. One first determines the e-ph matrix elements in the Bloch representation using the electronic and vibrational properties, computed with DFT and DFPT on coarse $\mathbf{k}$ - and $\mathbf{q}$ -point grids, respectively. The e-ph matrix element is defined as:

$$g_{mn\nu}(\mathbf{k}, \mathbf{q}) = \langle \psi_{m\mathbf{k}+\mathbf{q}} | \Delta_{\mathbf{q}\nu} V^{\text{KS}} | \psi_{n\mathbf{k}} \rangle, \quad (68)$$

where $\psi_{n\mathbf{k}}$ and $\psi_{m\mathbf{k}+\mathbf{q}}$ are the KS wavefunctions of the initial and final Bloch states (with $\mathbf{k}$ being the electron wavevector, and n being the band index), and $\Delta_{\mathbf{q}\nu} V^{\text{KS}}$ is the derivative of the self-consistent potential associated with a phonon with momentum $\mathbf{q}$ and branch index ν . The latter quantity can be obtained as:

$$\Delta_{\mathbf{q}\nu} V^{\text{KS}} = \sum_{\kappa\alpha p} e^{i\mathbf{q}\cdot\mathbf{R}_p} \frac{\partial V^{\text{KS}}}{\partial \tau_{\kappa\alpha p}} u_{\kappa\alpha, \mathbf{q}\nu}, \quad (69)$$

with

$$u_{\kappa\alpha,\mathbf{q}\nu} = \sqrt{\frac{\hbar}{2M_\kappa\omega_{\mathbf{q}\nu}}} e_{\kappa\alpha,\mathbf{q}\nu}. \quad (70)$$

Here $\mathbf{R}_p$ is the lattice vector identifying the unit cell p , $\tau_{\kappa\alpha p}$ is the position of atom κ in unit cell p in the Cartesian direction α , M_κ is the mass of atom κ , $\omega_{\mathbf{q}\nu}$ is the phonon frequency, and $e_{\kappa\alpha,\mathbf{q}\nu}$ is the eigendisplacement vector corresponding to atom κ in the Cartesian direction α for a collective phonon mode $\mathbf{q}\nu$.

Next, one finds the e-ph matrix elements in the Wannier representation:

$$g_{ij\kappa\alpha}(\mathbf{R}_e, \mathbf{R}_p) = \left\langle \mathbf{0}_e i \left| \frac{\partial V^{\text{KS}}}{\partial \tau_{\kappa\alpha p}} \right| \mathbf{R}_e j \right\rangle, \quad (71)$$

where $\mathbf{R}_e$ and $\mathbf{R}_p$ are the Bravais lattice vectors associated to the electron and phonon WS supercells, and $|\mathbf{R}_e j\rangle$ are the MLWFs with index j and centered in the cell at $\mathbf{R}_e$. This is done by transforming the e-ph matrix elements from the coarse BZ $(\mathbf{k}, \mathbf{q})$ grids into the corresponding real-space supercells $(\mathbf{R}_e, \mathbf{R}_p)$ as:

$$g_{ij\kappa\alpha}(\mathbf{R}_e, \mathbf{R}_p) = \frac{1}{N_e N_p} \sum_{\mathbf{k}, \mathbf{q}} e^{-i(\mathbf{k} \cdot \mathbf{R}_e + \mathbf{q} \cdot \mathbf{R}_p)} \times \sum_{mn\nu} V_{\mathbf{k}+\mathbf{q},im}^\dagger g_{mn\nu}(\mathbf{k}, \mathbf{q}) V_{\mathbf{k},nj} u_{\kappa\alpha,\mathbf{q}\nu}^{-1}. \quad (72)$$

In Eq. (72), N_e and N_p are the number of unit cells in the periodic BvK supercells corresponding to the number of $\mathbf{k}$ and $\mathbf{q}$ points on the coarse electron and phonon grids, respectively, and $V_{\mathbf{k}}$ is the Wannierization matrix introduced in Eq. (5). That matrix is provided by the Wannier engine, while $u_{\kappa\alpha,\mathbf{q}\nu}$ is obtained by diagonalizing the dynamical matrix at wavevector $\mathbf{q}$.

Finally, performing the inverse Fourier transform of Eq. (72), the e-ph matrix elements on very fine $(\mathbf{k}', \mathbf{q}')$ BZ grids are given by:

$$g_{mn\nu}(\mathbf{k}', \mathbf{q}') = \sum_{ep} e^{i(\mathbf{k}' \cdot \mathbf{R}_e + \mathbf{q}' \cdot \mathbf{R}_p)} \times \sum_{ij\kappa\alpha} \mathcal{U}_{\mathbf{k}'+\mathbf{q}',mi} g_{ij\kappa\alpha}(\mathbf{R}_e, \mathbf{R}_p) \mathcal{U}_{\mathbf{k}',jn}^\dagger u_{\kappa\alpha,\mathbf{q}'\nu}. \quad (73)$$

In this step, it is assumed that the e-ph matrix elements outside of the WS supercells defined by the initial coarse grids can be neglected. Prior to computing Eq. (73), the transformation matrices $\mathcal{U}_{\mathbf{k}',nj}$ given by Eq. (14) and the phonon eigenvectors $u_{\kappa\alpha,\mathbf{q}'\nu}$ for the new set of points $(\mathbf{k}', \mathbf{q}')$ must be found as described in Giustino *et al.* (2007a).

The accuracy of the Wannier-Fourier interpolation approach depends on the spatial localization of the $g_{ij\kappa\alpha}(\mathbf{R}_e, \mathbf{R}_p)$ matrix elements. Eq. (71) can be seen as a hopping integral between two localized WFs, one at $\mathbf{0}_e$ and one at $\mathbf{R}_e$, due to a perturbation caused by

the displacement of the atom at $\tau_{\kappa p}$. If the e-ph interactions are short-ranged in real space, the quantity $g_{ij\kappa\alpha}(\mathbf{R}_e, \mathbf{R}_p)$ decays rapidly with $|\mathbf{R}_e|$ and $|\mathbf{R}_p|$, and it is sufficient to only compute the matrix elements on a small set of $(\mathbf{R}_e, \mathbf{R}_p)$ lattice vectors to fully capture the coupling between electrons and phonons. As discussed in Giustino (2017) and Giustino *et al.* (2007a), the spatial decay is bound by the limiting cases $g_{ij\kappa\alpha}(\mathbf{R}_e, \mathbf{0}_p)$ and $g_{ij\kappa\alpha}(\mathbf{0}_e, \mathbf{R}_p)$. In the first case, the matrix element decays in the variable $\mathbf{R}_e$ at least as fast as the MLWFs. In the second case, the matrix element decays with the variable $\mathbf{R}_p$ at the same rate as the screened Coulomb potential generated by the atomic displacements. Thus, the localization of $g_{ij\kappa\alpha}(\mathbf{R}_e, \mathbf{R}_p)$ depends strongly on the dielectric properties of the system. In metals and non-polar semiconductors, the screening properties are dictated by Friedel oscillations (Fetter and Walecka, 2003) $|\mathbf{R}_p|^{-4}$ and quadrupole behavior (Pick *et al.*, 1970) $|\mathbf{R}_p|^{-3}$, respectively. In polar materials (i.e., ionic and polar covalent crystals), the dominant contribution to the potential is the dipole Fröhlich term (Vogl, 1976), which is long-ranged and decays as $|\mathbf{R}_p|^{-2}$. As a consequence, some of the matrix elements in reciprocal space diverge as $|\mathbf{q}|^{-1}$ in the long-wavelength limit ($\mathbf{q} \rightarrow 0$), and cannot be directly interpolated from a coarse to a fine grid using the Wannier-based interpolation approach. Recent methodologies have been developed for including long-range dipole (Deng *et al.*, 2021; Sjakste *et al.*, 2015; Sohler *et al.*, 2016; Verdi and Giustino, 2015) and quadrupole (Brunin *et al.*, 2020a,b; Jhalani *et al.*, 2020; Park *et al.*, 2020; Zhang and Liu, 2022) corrections in the interpolation of the e-ph matrix elements.

2. Codes

EPW (EPW, 2023; Lee *et al.*, 2023; Noffsinger *et al.*, 2010; Poncé *et al.*, 2016), the first open-source code for the study of e-ph interaction using MLWFs, was publicly released in 2010 and has been distributed within the Quantum ESPRESSO suite (Giannozzi *et al.*, 2017; Quantum ESPRESSO, 2023) since 2016. Several Wannier-based open-source codes exist today to compute physical properties related to e-ph interactions such as Perturbo (PERTURBO, 2023; Zhou *et al.*, 2021a), Phoebe (Cepellotti *et al.*, 2022; Phoebe, 2023), elphbolt (EPIq, 2023a; Protik *et al.*, 2022), and EPIq (EPIq, 2023b; Marini *et al.*, 2023). At present, EPW, Perturbo, Phoebe, and EPIq are all interfaced with Quantum ESPRESSO (Giannozzi *et al.*, 2017) to generate the relevant first-principles input data, and use Wannier90 (Pizzi *et al.*, 2020) in standalone or library mode to compute the required quantities in the Wannier representation. elphbolt, on the other hand, relies on EPW to generate the required Wannier space information. All codes follow an overall similar workflow to compute

e-ph matrix elements on fine grids, as outlined below and summarized in Fig. 7.

1. The initial step is to perform DFT calculations with **Quantum ESPRESSO** on a uniform coarse electronic $\mathbf{k}$ -point grid to obtain the band energies and Bloch wavefunctions, and DFPT calculations with **Quantum ESPRESSO** on an irreducible coarse $\mathbf{q}$ -point grid to obtain the dynamical matrices and the derivatives of the self-consistent potential with respect to the phonon perturbations. **Phoebe** and **EPIq** require that the e-ph matrix elements on the coarse electron and phonon grids are also computed with **Quantum ESPRESSO** since these quantities are later passed to the two codes. **EPW** and **PERTURBO**, on the other hand, compute the e-ph matrix elements on the coarse $\mathbf{k}$ - and $\mathbf{q}$ -point grids internally by reading the files generated from the DFT and DFPT calculations.

2. Next, one must perform a precise Wannierization of the system using **Wannier90** in standalone mode with **PERTURBO**, **Phoebe**, and **EPIq**, or in library mode with **EPW**. This step produces the MLWFs and the Wannierization matrices $V_{\mathbf{k}}$ that transform the DFT Bloch wavefunctions into MLWFs.

3. The next step is to compute the e-ph matrix elements on the coarse $\mathbf{k}$ - and $\mathbf{q}$ -point grids in the Bloch representation and transform them, along with the electronic Hamiltonian and the dynamical matrix, from the Bloch to the Wannier representation. As noticed at point 2, **Phoebe** and **EPIq** uses the e-ph matrix elements on the coarse electron and phonon grids computed directly with **Quantum ESPRESSO**.

4. The final step is to perform an inverse Fourier transform of the electronic Hamiltonian, dynamical matrix, and e-ph matrix elements from the Wannier to the Bloch representation. At this stage, the electronic eigenvalues, phonon frequencies, and e-ph matrix elements can be efficiently computed on ultra-dense $\mathbf{k}'$ - and $\mathbf{q}'$ -point grids and further used to carry out calculations of various materials properties.

To extend the Wannier-based interpolation scheme to systems with long-range e-ph contributions, the strategy is to split the e-ph matrix elements into a short- and long-range part. Then, the short-range e-ph matrix elements are interpolated to ultra-dense $\mathbf{k}'$ - and $\mathbf{q}'$ -point grids using MLWFs, while the long-range e-ph matrix elements are computed using derived first-principles expressions for the dipole and quadrupole e-ph interactions. Finally, the short- and long-range components are summed up to recover the total e-ph matrix elements.

A wide range of properties can be currently computed with **EPW**, **Perturbo**, **Phoebe**, **elphbolt**, and **EPIq**. A full list of the capabilities for the released version of each code can be found on the respective website. In particular, **EPW** computes phonon-limited transport properties in the framework of the BTE, magnetotransport, phonon-mediated superconducting properties based on the Migdal-Eliashberg formalism, phonon-assisted op-

tical processes, and polarons (**EPW**, 2023; Lee *et al.*, 2023). **EPW** also comes with the **ZG** toolkit for calculations of temperature-dependent e-ph properties for harmonic and anharmonic phonons via the special displacement method (Zacharias and Giustino, 2016, 2020; Zacharias *et al.*, 2023a,b). **Perturbo** calculates phonon-limited transport properties using BTE, ultrafast carrier dynamics, magnetotransport, and high-field transport (**PERTURBO**, 2023; Zhou *et al.*, 2021a). **Perturbo** is interfaced with **TDEP** (**TDEP**, 2023) to compute anharmonic phonons. **Phoebe** provides various tools to predict electron and phonon transport properties at different levels of theory and accuracy, such as the BTE or models based on the Wigner distribution (Cepellotti *et al.*, 2022; Phoebe, 2023). **elphbolt** solves the coupled electron and phonon BTEs, and the effect of the mutual e-ph drag on the electrical and thermal transport coefficients (**EPIq**, 2023a; Protik *et al.*, 2022). **EPIq** computes phonon-mediated superconducting properties based on the Migdal-Eliashberg formalism, double resonant Raman intensities, and excited carriers lifetimes (**EPIq**, 2023b; Marini *et al.*, 2023).

H. Beyond DFT with localized orbitals

While in most cases DFT is the method of choice for electronic ground state calculations, and even the ground state properties of certain materials, may require a “beyond-DFT” treatment for accurate first-principles predictions. Two examples are finite-temperature and spectroscopic properties, as observed in direct and inverse photoemission experiments, which cannot be addressed adequately within conventional DFT. Similarly, the complex physics arising from strong local Coulomb interaction in partially filled orbitals is beyond the scope of a single-particle picture, which can manifest itself in an inaccurate description of the material. In such cases, more advanced methods are needed. One class of approaches is based on diagrammatic many-body perturbation theory; examples include the GW approximation or DMFT (see Sec. III.H.1). Since such methods are often computationally costly and complex, it may be necessary to extract accurate low-energy effective Hamiltonians that are treated using these methods in a post-processing step. An efficient alternative is to retain the functional character of DFT and apply physically motivated corrections, as in hybrid or Koopmans-compliant functionals (see Sec. III.H.2). In this case the theory is no longer based on a pure functional of the density, but the orbitals themselves or their orbital densities become the key variables. Common to both approaches is the importance of improving the description of local, orbital-dependent physics. This is where WFs come in, providing a useful basis for such applications and supporting the physical understanding with chemical intuition.

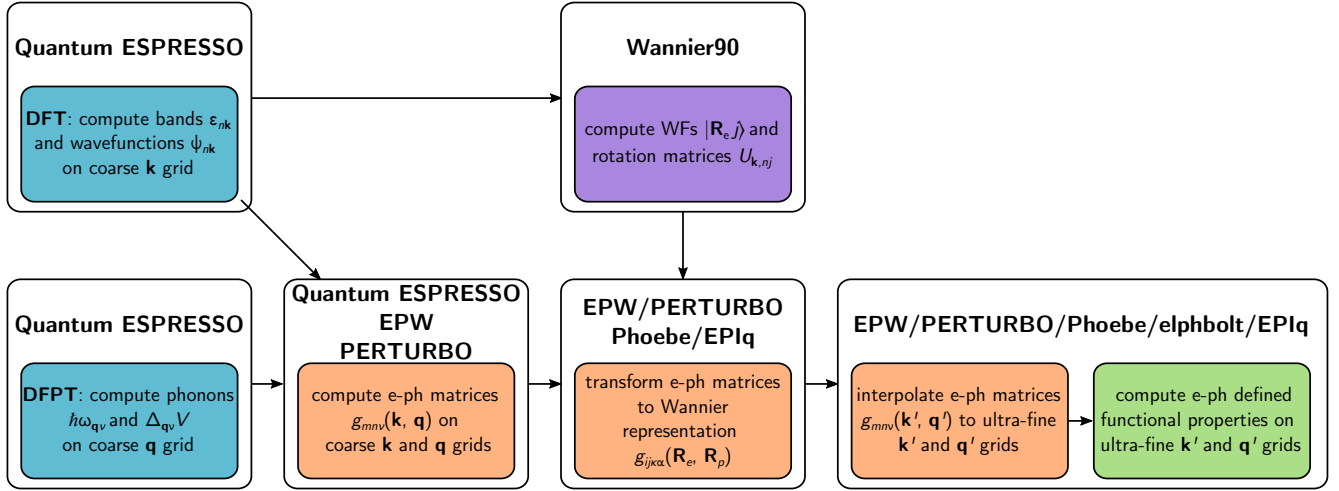


FIG. 7 Workflow to compute e-ph interactions on ultra-fine grids using WFs. The user performs DFT and DFPT calculations on coarse $\mathbf{k}$ and $\mathbf{q}$ -point grids with Quantum ESPRESSO. Next, the user computes the e-ph matrix elements on the coarse grids and transforms them in localized Wannier basis using the rotation matrices obtained with Wannier90. Note that Phoebe and EPIq read the e-ph matrix elements on the coarse grids computed with Quantum ESPRESSO, while EPW and PERTURBO compute them internally. Finally, EPW/PERTURBO/Phoebe/EPIq interpolate the band structure, phonon dispersion, and e-ph matrix elements on ultra-fine $\mathbf{k}'$ - and $\mathbf{q}'$ -point grids, and perform calculations of various materials properties.

1. Dynamical mean-field theory and embedding

Strongly correlated electron systems host a wide variety of physical phenomena, ranging from Mott physics to high-temperature superconductivity to exotic ordered phases (Tokura *et al.*, 2017). Fundamental to these phenomena is the competition of itinerant versus localized character of electrons, which requires computational methods beyond the single-particle picture. While the representation of electronic states in reciprocal space can be beneficial, the theory of strong local correlations is most naturally formulated in a real-space basis. A key aspect of many beyond-DFT schemes is therefore the combination of itinerant Bloch states with localized molecular orbitals, which can be elegantly formalized with WFs (Lechermann *et al.*, 2006).

Beyond-DFT schemes are typically computationally demanding and require a compromise in terms of the number of orbitals that can be treated. Starting from an *ab initio* description of a large number of bands, it is common practice to select a subset of correlated orbitals, such as those describing electronic states in the vicinity of the Fermi level, using projector functions. The result is a so-called downfolded model, which contains only the relevant degrees of freedom (Kotliar *et al.*, 2006). It divides the total system into a subspace of localized orbitals, for which a higher-level method is used, and the remaining KS states, for which the single-particle description within DFT accurately reflects the physics. The embedding *ansatz* is further justified by the observation that correlated physics phenomena typically occur on energy scales of meV to a few eV (Chen *et al.*, 2022). For the

remainder of this subsection we focus on DMFT as the higher-level method to solve the downfolded many-body problem, but the concepts can be similarly applied to its extensions (see below) and conceptual other approaches (see, e.g., Eskridge *et al.* (2019); Muechler *et al.* (2022); Sheng *et al.* (2022); and Zgid and Gull (2017)).

While the multiband Hubbard model studied with DMFT is naturally related to TB models, a rigorous formalism based on energy functionals allows the combined approach with *ab initio* methods, as in DFT+DMFT (Kotliar *et al.*, 2006). The development of DFT+DMFT methods and software for strongly correlated materials has seen a tremendous surge in the last few decades (Held, 2007; Paul and Birol, 2019; Pavarini *et al.*, 2011). Routine calculations allow to compute single-particle spectra (Anisimov *et al.*, 2005; Nekrasov *et al.*, 2006; Pavarini *et al.*, 2004), optical conductivity (Haule *et al.*, 2005; Wissgott *et al.*, 2012), transport (Oudovenko *et al.*, 2006; Zingl *et al.*, 2019) and thermoelectric properties (Arita *et al.*, 2008; Tomczak *et al.*, 2012; Wissgott *et al.*, 2010), electronic Raman (Blesio *et al.*, 2023), and two-particle correlation functions (susceptibilities) (Boehnke *et al.*, 2011; Kuneš, 2011; Park *et al.*, 2011). Recent advances include interactions with core holes as in x-ray absorption and photoemission spectroscopy, and resonant inelastic x-ray scattering (Hariki *et al.*, 2017, 2018; Haverkort *et al.*, 2012; Lüder *et al.*, 2017). Furthermore, routines for lattice optimization (Haule and Pascut, 2016; Leonov *et al.*, 2014; Plekhanov *et al.*, 2021) and for computing phonon spectra (Koçer *et al.*, 2020) have been formalized. This list of references is by no means exhaustive, but is intended

to serve as a starting point for the respective topics.

In beyond-DFT methods, it is convenient to describe the total system of interacting electrons in a periodic solid in terms of the momentum- and frequency-dependent retarded single-particle Green's function

$$\hat{G}(\mathbf{k}, \omega) = \left[(\omega + \mu)\mathbb{1} - \hat{H}(\mathbf{k}) - \hat{\Sigma}(\mathbf{k}, \omega) + i\eta \right]^{-1}, \quad (74)$$

where μ is the chemical potential and $\hat{H}(\mathbf{k})$ represents the non-interacting Hamiltonian. The frequency- and momentum-dependent electron self-energy is given by $\hat{\Sigma}(\mathbf{k}, \omega)$, and η is an infinitesimal positive parameter to ensure physical correctness. For clarity, we have omitted the double-counting correction here, and we refer to Karolak *et al.* (2010) for an overview. Starting from a DFT-derived downfolded Hamiltonian, the challenge is to compute the corresponding self-energy correction that accounts for dynamical interaction effects. Various approaches can be formalized, but for the purpose of this review we outline the workflow of single-site DMFT (Georges *et al.*, 1996). In DMFT, the self-energy becomes a site-local quantity for a given atomic site $\mathcal{R}$ within the unit cell when expressed in a localized orbital basis. This approximation is conceptually similar to DFT+ U (Anisimov *et al.*, 1997), but in DMFT the full frequency dependence of the interaction is taken into account. Following this approach, in the DMFT self-consistency cycle, the local lattice self-energy $\Sigma^{\mathcal{R}}(\omega)$ is approximated by that of an auxiliary quantum impurity problem. The most computationally challenging step of the DMFT loop is typically to find the solution to the impurity problem, which allows to infer the impurity self-energy via the Dyson equation. The self-energy is embedded into the Hilbert space of the effective Hamiltonian as

$$\Sigma_{mn}(\mathbf{k}, \omega) = \sum_{\mathcal{R}, ij} P_{mi}^{\mathcal{R}*}(\mathbf{k}) \Sigma_{ij}^{\mathcal{R}}(\omega) P_{jn}^{\mathcal{R}}(\mathbf{k}). \quad (75)$$

Approximating the lattice self-energy in Eq. (74) by the upfolded impurity self-energy becomes exact for infinite connectivity of the lattice (Georges and Kotliar, 1992; Metzner and Vollhardt, 1989). The projector functions $P_{jn}^{\mathcal{R}}(\mathbf{k})$ in (75) encode the basis transformation from band to orbital basis, i.e., from $|\psi_{n\mathbf{k}}\rangle$, with band index n and wavevector $\mathbf{k}$, to $|\psi_{\mathcal{R}j\mathbf{k}}^{\mathcal{W}}\rangle$, with orbital index j at site $\mathcal{R}$ (i.e., j is an intrasite index here):

$$P_{jn}^{\mathcal{R}}(\mathbf{k}) = \langle \psi_{\mathcal{R}j\mathbf{k}}^{\mathcal{W}} | \psi_{n\mathbf{k}} \rangle. \quad (76)$$

The local Green's function is then computed as $G_{ij}^{\text{loc}, \mathcal{R}}(\omega) = \frac{1}{N} \sum_{\mathbf{k}, mn} P_{im}^{\mathcal{R}}(\mathbf{k}) G_{mn}(\mathbf{k}, \omega) P_{nj}^{\mathcal{R}*}(\mathbf{k})$, where N is the total number of k -points of the grid. To determine a suitable localized basis set, some DFT+DMFT codes use projections on atomic orbitals, others rely on Wannier90 directly for a simple and user-friendly interface. While the two approaches are conceptually similar

(for a more detailed overview see Chen *et al.* (2022)), the choice of projectors may affect the results and therefore needs to be carefully analyzed (Karp *et al.*, 2021). Note that Eq. (76) assumes that the DFT+DMFT calculation is performed in the band basis in a charge self-consistent mode (i.e., $P(\mathbf{k})$ corresponds to $V_{\mathbf{k}}$ in Eq. (5b)). However, for one-shot calculations, the equations simplify (Beck *et al.*, 2022). Wannier interpolation can be used in the DMFT self-consistent loop for an isolated set of bands or at the TB level, which is crucial for accurately resolving low-energy physics (Kaye *et al.*, 2023).

Multiple schemes go beyond standard DMFT, but the discussion above carries over directly to these extensions. Examples include cluster-DMFT approaches (either in real or reciprocal space) or diagrammatic extensions of the self-energy such as the DFA (Galler *et al.*, 2017, 2019) and the TRILEX method (Aryal and Parcollet, 2015, 2016), as well as non-equilibrium DMFT (Aoki *et al.*, 2014). To improve some of the shortcomings of the DFT+DMFT method, a better starting point than DFT may be GW, which combined with DMFT allows to include non-local effects beyond DFT as well as to formalize the double-counting correction term (Biermann, 2014).

a. DFT+DMFT codes From the previous section, it becomes clear that a fully integrated DFT+DMFT software suite requires three main components: 1) a DFT implementation (*ab initio* engine) and a routine to construct the localized basis set (e.g., a Wannier engine), 2) a Green's function formalism to implement the DMFT equations, and 3) an impurity solver. At present, there are several open-source implementations that meet these requirements to varying degrees. On the side of more monolithic, publicly available beyond-DFT codes, there are implementations in, for example, CASTEP (Plekhanov *et al.*, 2018), Abinit (Romero *et al.*, 2020), RSPt (Di Marco *et al.*, 2009; Grechnev *et al.*, 2007; Thunström *et al.*, 2009), Amulet (AMULET, 2023), and eDMFT (Haule *et al.*, 2010), which include an implementation of DFT and a downfolding routine, as well as choices of internal and externally linked impurity solvers. ComDMFT (Choi *et al.*, 2019), on the other hand, interfaces directly with Wannier90 for the downfolding procedure. All codes support charge self-consistency.

An alternative philosophy is a more modular library approach, focusing on providing the framework for performing DMFT calculations based on input from a DFT calculation. For this purpose, most codes directly rely on Wannier90 to benefit from a generic, robust, and flexible interface independent of the flavor of DFT. Examples include w2dynamics (Wallerberger *et al.*, 2019) and DCORE (Shinaoka *et al.*, 2021), as well as EDIPack (Amaricci *et al.*, 2022), DMFTwDFT (Singh *et al.*, 2021), and

TRIQS (Aichhorn *et al.*, 2016; Merkel *et al.*, 2022; Parcollet *et al.*, 2015), with only the latter two packages supporting charge self-consistency at present (Beck *et al.*, 2022; James *et al.*, 2021; Schüler *et al.*, 2018; Singh *et al.*, 2021). All packages contain a number of internal and external impurity solvers. A list of all currently available impurity solvers is beyond the scope of this article, but can be found on the respective websites.

b. Interaction-parameters codes As in DFT+ U , the local interaction parameters that enter the Hubbard model in DFT+DMFT must be chosen appropriately. Starting from a local basis set, the Coulomb integrals can be evaluated, ideally taking into account screening processes and the symmetries of the system to simplify the parametrization of the interaction Hamiltonian (Chen *et al.*, 2022). The most widely used approach is the constrained Random Phase Approximation (cRPA) (Aryasetiawan *et al.*, 2004). Currently, the method is implemented in ABINIT (Romero *et al.*, 2020), SPEX (Friedrich *et al.*, 2010), VASP (Kaltak, 2015), and RESPACK (Nakamura *et al.*, 2021), with the latter three offering the possibility to use WFs constructed by Wannier90. Alternatively, the interaction parameters are treated as free parameters that must be adjusted to match experimental observables.

c. Post-processing Once the solution to the DFT+DMFT scheme is found (charge) self-consistently (see Fig. 8), converged DMFT quantities such as the self-energy, the local Green’s function, or the hybridization function allow to compute several physical observables in post-processing applications. Depending on the frequency domain in which the impurity solver operates, it may be necessary to use analytic continuation to obtain the self-energy in the real frequency domain (Gubernatis *et al.*, 1991; Wang *et al.*, 2009). Such programs are often included in the respective software packages. Since the post-processing step is performed only once after convergence, and therefore does not significantly contribute to the overall computational cost, Wannier interpolation (see Sec. III.C) is particularly beneficial at this stage. A standard observable is the lattice or impurity spectral function, which is directly related to photoemission and absorption spectra. Most software packages provide tools for users to compute such quantities routinely. Transport tensors based on Kubo’s linear response theory (Kubo, 1957), such as optical and thermal conductivities, as well as Hall and Seebeck coefficients, can be computed using various tools, including TRIQS/DFTTools (Aichhorn *et al.*, 2016) and the woptic package (Assmann *et al.*, 2016). Another option is LinReTraCe (Pickem *et al.*, 2023), which relies on a semi-analytical approach valid when

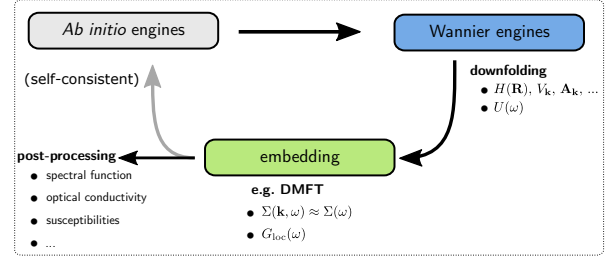


FIG. 8 Typical workflow of the embedding formalism. First, the user performs an *ab initio* calculation from which a downfolded model is derived in the basis of localized orbitals. The downfolded model is solved using an appropriate many-body method such as DMFT (depicted). Physical observables can be computed in a post-processing step. For full self-consistency the cycle is iterated until convergence.

a linear expansion of the self-energy is adequate. Tools to evaluate core level spectroscopies within different levels of approximations are available, for example, in Quanty (Haverkort, 2016) and EDRIXS (Wang *et al.*, 2019b).

2. Koopmans functionals

Koopmans-compliant (KC) functionals (Borghi *et al.*, 2014; Colonna *et al.*, 2022, 2018, 2019; Dabo *et al.*, 2010; De Gennaro *et al.*, 2022; Elliott *et al.*, 2019; Nguyen *et al.*, 2018) are orbital-dependent functionals capable of delivering accurate spectral properties for molecular and extended systems at low computational cost. Remarkably, the KC approach maintains a simple functional formulation while being more accurate than G_0W_0 and comparable to quasi-particle self-consistent GW with vertex correction ($QSGW$) (Colonna *et al.*, 2022, 2019; Nguyen *et al.*, 2018), at a cost which is broadly comparable to standard DFT. The simplicity and accuracy of the KC framework rests on three fundamental concepts: linearization, screening, and localization. First, a generalized linearization condition is imposed on each charged excitation: the energy of any orbital must be independent of the occupation of the orbital itself. This implies that the KC total energy functional is piecewise linear with respect to fractional occupations, and essentially implements a generalized definition of self-interaction-free orbital. Second, electronic screening and orbital relaxation (due to the electron addition/removal process) are taken into account by orbital-dependent screening coefficients, which can be calculated by finite differences (Nguyen *et al.*, 2018) or linear-response approaches (Colonna *et al.*, 2022, 2018). Finally, the Koopmans compliance condition is imposed on those variational orbitals—i.e., those minimizing the KC energy functional—which are localized. For periodic systems, these variational orbitals resemble MLWFs (Colonna *et al.*, 2022, 2018, 2019;

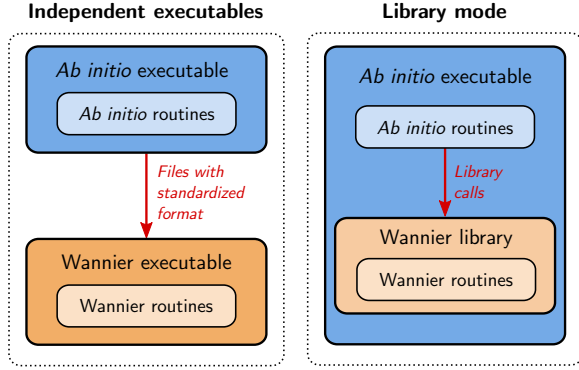


FIG. 9 Different approaches for interaction of the *ab initio* codes with the library routines. Left: the *ab initio* codes and the Wannier engine are compiled independently in two different executables. Data exchange happens via files with standardized formats (see also discussion in Secs. III.B and III.I.2). Right: the Wannier engine provides a library mode. The *ab initio* code is linked at compile time to the Wannier library, and a single executable is created. The *ab initio* routines call directly the routines from the Wannier engine via library calls.

(Nguyen *et al.*, 2018).

Using WFs as a proxy for variational orbitals has allowed the development of a Wannier-interpolation and unfolding scheme to calculate the band structure from a supercell Koopmans-functional calculation (De Gennaro *et al.*, 2022). In addition, WFs have fostered the development of a convenient Koopmans formulation that operates fully under PBCs, and is based on explicit BZ sampling and DFPT (Colonna *et al.*, 2022). This *Koopmans-Wannier* implementation, which goes under the name of KCW, is available in the Quantum ESPRESSO distribution; it delivers improved scaling with system size, and makes band-structure calculations with Koopmans functionals straightforward. KC functionals resonate with other efforts aimed at calculating excitation energies where WFs and localized orbitals are often a key ingredient (Anisimov and Kozhevnikov, 2005; Anisimov *et al.*, 2007; Kraisler and Kronik, 2013; Li *et al.*, 2015, 2017; Ma and Wang, 2016; Skone *et al.*, 2014; Wing *et al.*, 2021).

I. Interoperability between codes in the ecosystem

1. Library mode for the Wannier engines

When combining two (or more) codes, various approaches are possible. While the most common approach has been to compile the Wannier engine into a different executable than the *ab initio* code, with data being transferred via files with standardized formats, an alternative approach is to expose the Wannier routines via a library interface, as schematically depicted in Fig. 9. In this

second approach, a single executable is created, and the main *ab initio* code is responsible for calling the appropriate routines from the Wannier-engine library.

Making sure that the library interface is both easily usable and covers all possible use cases, however, is a non-trivial task. In the specific case of Wannier90, for instance, its first release contained a simple interface to allow it to be called as a library from another Fortran program, with the necessary data being passed by the calling program rather than by file. Over time, however, it became apparent that this original library interface did not provide the full functionality needed by an ecosystem of codes, presenting three main issues. First, the interface was not fully compatible with parallel calling codes, because there was no means to distribute the data and make use of Wannier90’s internal parallelism. Second, the use of global module variables meant that Wannier90 was not thread-safe, i.e., a calling program could not call more than one instance of Wannier90. The third issue concerns error handling. On detecting an error, Wannier90 would abort, causing the calling program to crash. The desired behavior would be instead to return an error code, allowing the calling program to decide how to handle it (for instance, exiting gracefully or retrying with different parameters). Work to address all three of these issues has been carried out by a team at the Scientific Computing Department of STFC at Daresbury Laboratory, UK, in collaboration with three of the present authors (AAM, JRY and GP). A development version is available in the Wannier90 GitHub repository and will be merged into a future Wannier90 release.

2. File I/O generation and parsing

In the first approach mentioned earlier, intermediate files (as listed in Tab. I) are used to decouple the first-principles calculations from the Wannierization step. As already discussed earlier, the formats defined by Wannier90 became the *de facto* standards for the codes in the Wannier-functions software ecosystem. For instance, many TB codes can read the so-called `.tb.dat` or `.hr.dat` files (as already mentioned, these contain on-site energies, hopping terms, ...) to further process the Wannier Hamiltonian. In the past few years many software packages have thus started to implement their own parsers for these I/O files, often focusing only on a specific subset of the file formats relevant for their own use case. However, this leads to a duplication of efforts, also considering that maintaining robust and feature-complete parsers is a non-trivial task. As many of these codes use Python as the programming language of choice, such as TBmodels (TBmodels, 2023), PythTB (PythTB, 2023a), WannierBerri (Tsirkin, 2021), and AiiDA (Huber *et al.*, 2020; Pizzi *et al.*, 2016; Uhrin *et al.*, 2021), a community effort has been initiated by Jamal I. Mustafa

and others, to implement a centralized reference set of parsers for the Wannier90 input/output files in Python, hosted at <https://github.com/wannier-developers/wannier90io-python/>. The goal of this project is not only to provide a parser library for developers, but also to provide a convenient package directly for users, allowing them to easily load and manipulate the Wannier90 input/output files for their own use case.

In addition, part of the Wannier90 code base already outputs Python scripts for post-processing; one example is the `berry` module, which outputs scripts using the `matplotlib` (Hunter, 2007) library to plot the Berry curvature. However, these scripts are hard-coded into the Wannier90 codebase as a series of `write` Fortran statements that output Python code, which makes them very difficult to update and maintain. The aforementioned Python library will also facilitate future efforts on moving these post-processing functions into a dedicated Python package, in the hope of smoothening the development experience, as well as allowing users to easily postprocess and visualize the calculation results.

The concept of a common parsing library is not limited to the Python language, and can also be applied to other emerging languages. For example, the Julia package `WannierIO.jl` (Qiao *et al.*, 2023c) provides functions to read/write Wannier90 file formats, and is used by the `Wannier.jl` (Qiao *et al.*, 2023d) and `DFTK.jl` (Herbst *et al.*, 2023) packages as their I/O backend.

J. Automation, workflows and high-throughput

The diversity of the software ecosystem demonstrates the effectiveness of WFs. However, all methods face the need of a robust Wannierization, which used to be a not very straightforward process, since it involves a series of DFT calculations and construction of WFs, and more importantly it also depends sensitively on various input parameters (number of WFs, initial projections for ML-WFs, energy windows, k -point sampling, etc.). Their selection often required experience and chemical intuition, and was often a major challenge not only for beginners, but even for experienced researchers. Fully automated Wannierizations would make the procedure straightforward, and, as a consequence, allow any researcher to easily use all capabilities of the whole ecosystem, while also enabling high-throughput studies for accelerated materials discovery. To this end, it became urgent and necessary to perform algorithmic developments on the Wannierization itself, and to implement robust workflows combining multiple software packages in the ecosystem. On the algorithmic side, Wannierization should provide well-localized WFs without user input (for initial projections or energy windows, for instance); on the workflow side, one would like to orchestrate all different steps from the initial DFT calculations to the Wannier-engine exe-

cutions to the post-processing steps, while dynamically parsing the outputs and generating new inputs. Moreover, the workflow engine should provide a set of well-tested convergence parameters, and it should be able to handle common errors, and to automatically restart failed calculations.

Recent development of novel algorithms have largely solved the Wannierization challenge, starting from the “selected columns of the density matrix” SCDM (Damle and Lin, 2018; Damle *et al.*, 2015, 2017) algorithms, which generate initial projections by decomposing the density matrix, to the “projectability disentanglement” PDWF (Qiao *et al.*, 2023b), that uses projectability thresholds on atomic orbitals, rather than energy windows, to select which states to drop, keep frozen, or throw in the disentanglement algorithm. Together with the “manifold remixing” MRWF (Qiao *et al.*, 2023a) using parallel transport (Gontier *et al.*, 2019) these approaches remove what had been up to now a critical stumbling block.

On the workflow-engine side, several software packages are able to automate the electronic-structure calculations, such as `pymatgen` (Ong *et al.*, 2013) and `FireWorks` (Jain *et al.*, 2015), `AFLow $\pi$` (Supka *et al.*, 2017), `mkite` (Schwalbe-Koda, 2023), `ASE` (Larsen *et al.*, 2017) and `ASR` (Gjerding *et al.*, 2021), and `AiiDA` (Huber *et al.*, 2020; Pizzi *et al.*, 2016; Uhrin *et al.*, 2021); some of them, such as `ASE` and `AiiDA`, also provide functionalities or workflows to compute WFs. Equipped with automated Wannierization algorithms and robust workflow engines, it has become now possible to create workflows for automated Wannierizations. For instance, Gresch *et al.* (2018) implemented `AiiDA` workflows and gathered Wannier TB models for a group of III-V semiconductor materials; Vitale *et al.* (2020) used SCDM algorithm together with `AiiDA` workflows, carefully tested convergence parameters, and benchmarked Wannier interpolation accuracy on a set of 200 structures for entangled bands and a set of 81 structures for isolated bands; Garrity and Choudhary (2021) created a database of Wannier Hamiltonians for 1771 materials; Fontana *et al.* (2021) implemented workflows in `ASE` and Wannierized 30 inorganic monolayer materials using an automated protocol; and finally Qiao *et al.* (2023b) used PDWF to automate the Wannierization, obtaining over 1.3 million MLWF for over 20,000 3D inorganics from the Materials Cloud (Talirz *et al.*, 2020) MC3D database, then using manifold remixing (Qiao *et al.*, 2023a) to separately Wannierize back these into the valence and conduction bands of 77 insulators. These high-throughput studies can not only expedite materials discoveries, but also help identify challenging cases for the Wannierization algorithm, and promote further development of robust and automated Wannierization approaches.

IV. Conclusions and perspectives

The WFs software ecosystem represents a positive model for interoperability and decentralized code development in electronic-structure simulations, very much in the spirit of CECAM’s Electronic Structure Library (Oliveira *et al.*, 2020). This was made possible both by the nature of the scientific problem and physical quantities involved, and by the design choices originally made in **Wannier77** and **Wannier90**, planned early on as Wannierization engines decoupled from the *ab initio* codes used to compute the electronic structure. The availability of a well-documented, maintained, and modular open-source Wannier engine has pushed scientists to extend **Wannier90**’s functionalities or, when deemed more practical or efficient, to develop novel packages targeting specific materials properties. The growing availability of post-processing features has ignited a positive loop which further attracted developers from different electronic structure domains to work and use WFs, strengthening the interest in WF-related methods and resulting in the current ecosystem of interoperable software. The ecosystem has been reinforced by the organization of coding weeks and developer workshops, that have proven to be crucial to keep the community engaged and synced, to avoid duplication of efforts, and to collaborate on code maintenance.

While we could not cover here all the existing applications and codes leveraging WFs, we have outlined some of the most popular applications and summarized how they can be implemented in software packages and workflows to calculate advanced materials properties.

Looking forward, we expect that the ongoing efforts in the redesigned **Wannier90**’s library mode will be instrumental in smoothly integrating automated Wannierization procedures within *ab initio* and post-processing codes, with the benefit of reduced file I/O and code maintenance. As Wannierization becomes increasingly automated, we expect scientists to focus on calculations of complex properties, either through high-level programming of simulation workflows, or through the development and extension of post-processing packages. As a result, even more materials properties will become computationally accessible thanks to WFs, and available to the community through the release of dedicated functionalities, either in existing or in new packages of the ecosystem.

Finally, it is worth commenting on two crucial features of an ecosystem, being it biological or software: biodiversity and resilience. A certain level of biodiversity within a software ecosystem, i.e., the existence of multiple software packages with partially overlapping functionalities, can increase its robustness. First, it enables cross-verification of different implementations, increasing the reliability of the results and facilitating a rapid identification of bugs. Second, it can ensure that the

ecosystem capabilities are not lost if a package goes unmaintained or disappears. This aspect is connected to resilience, i.e., the capability of the ecosystem to deliver functionalities—such as the calculation of materials properties—under the loss of some of its components, an especially relevant issue in a scientific community where developers might not be able to guarantee long-term support for their code. We highlight that a software ecosystem might display the same dynamics that can be seen in biological settings, including competition and extinction. While a certain level of competition can result in code improvements regarding feature coverage, efficiency, and robustness, we caution that extreme competition might undermine biodiversity. It is thus important to sustain the work of individual developers who contribute to the progress and maintenance of an active, heterogeneous, and efficient ecosystem, encouraging measures to ensure proper scientific recognition. More broadly, the challenge will be to support the software development work, which is crucial for the long-term maintenance and integration of heterogeneous software packages.

We believe that a diverse, resilient, and open WF software ecosystem is a major asset for the electronic structure community in its quest to understand, discover, and design materials.

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